#14Days14Chapters

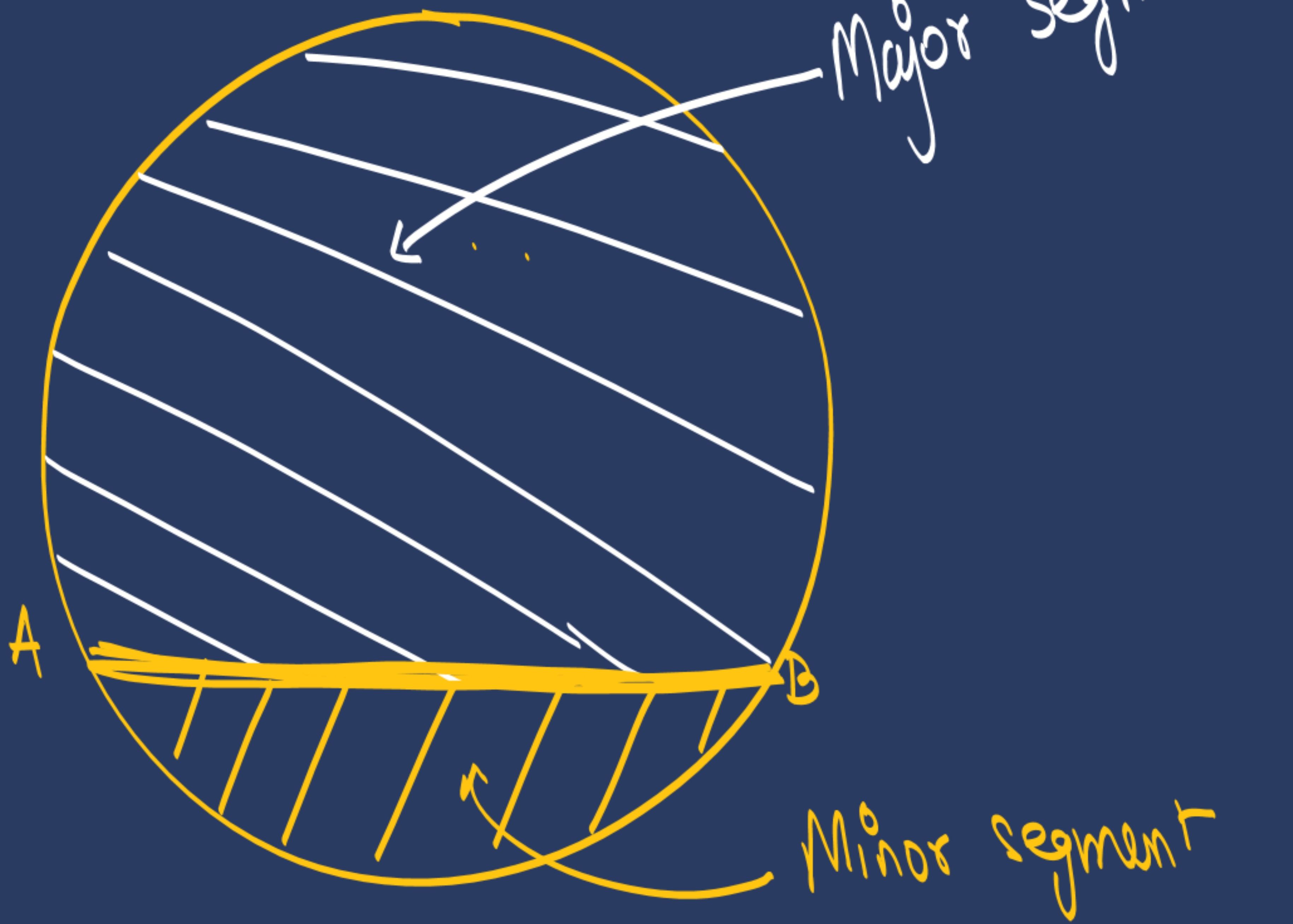
DAY 5

6,7,8 → 5 PM

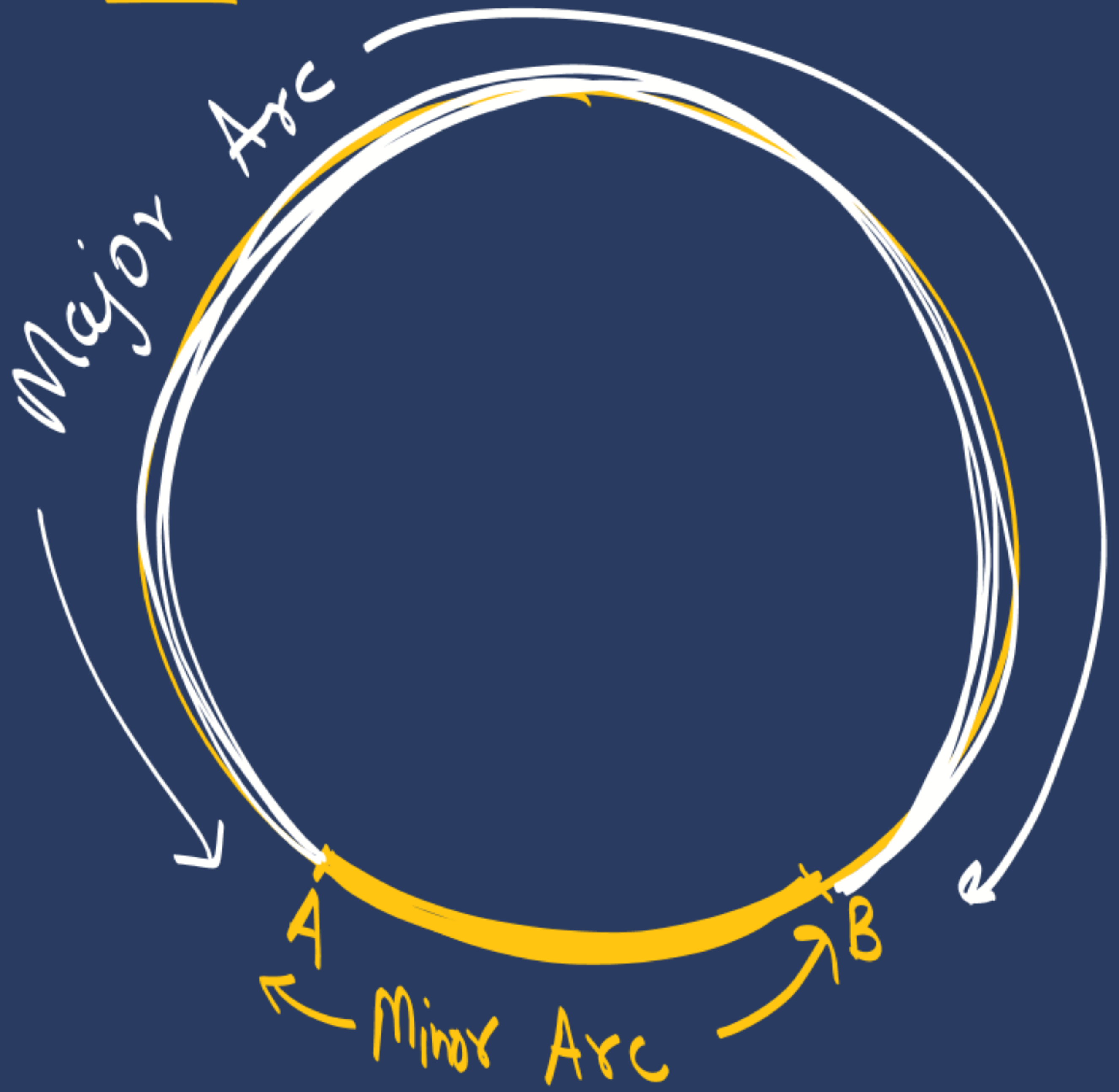
[Aim: 100/100 in Maths]

AREAS RELATED TO CIRCLES

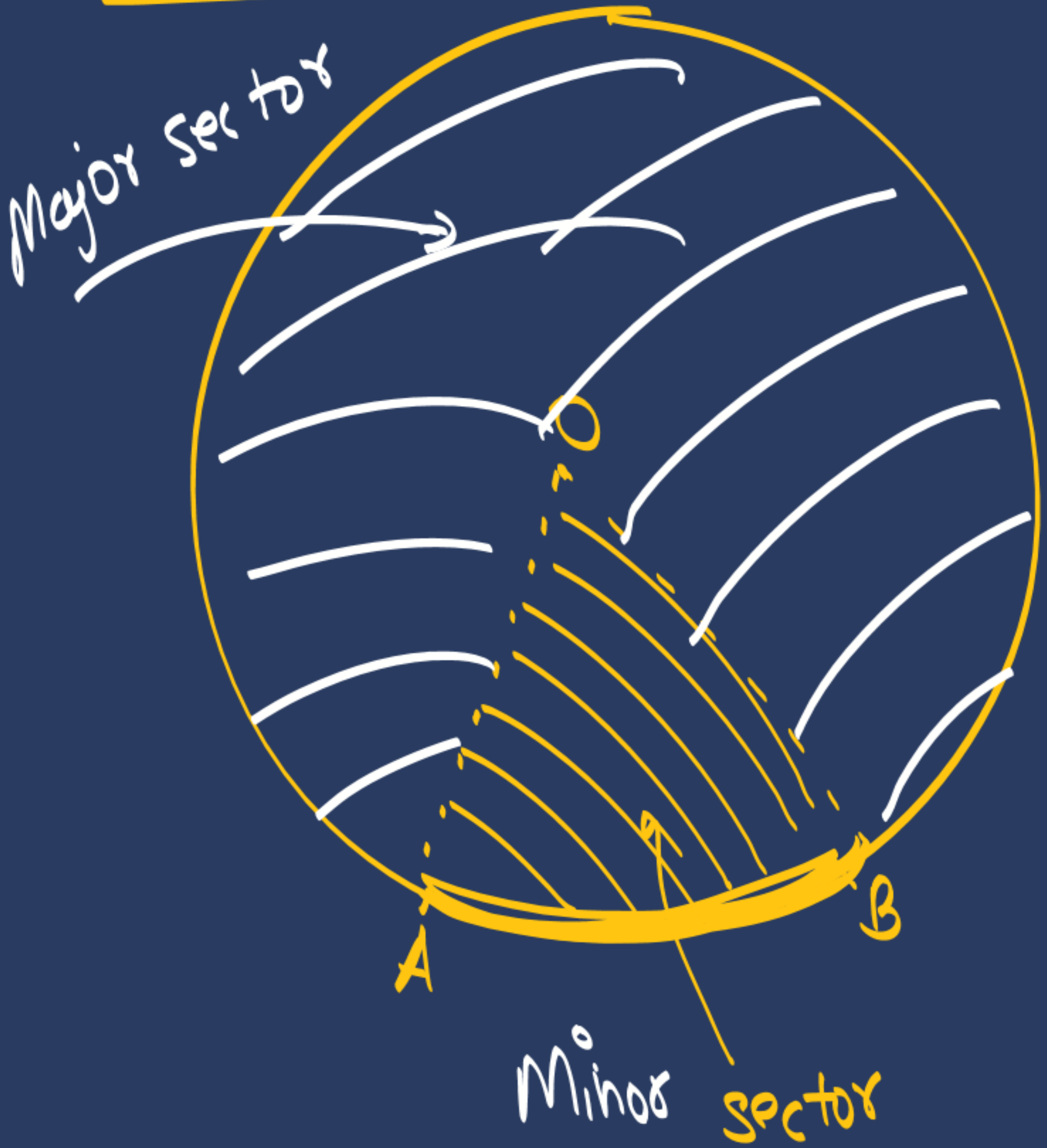
Segment :- (भाग)



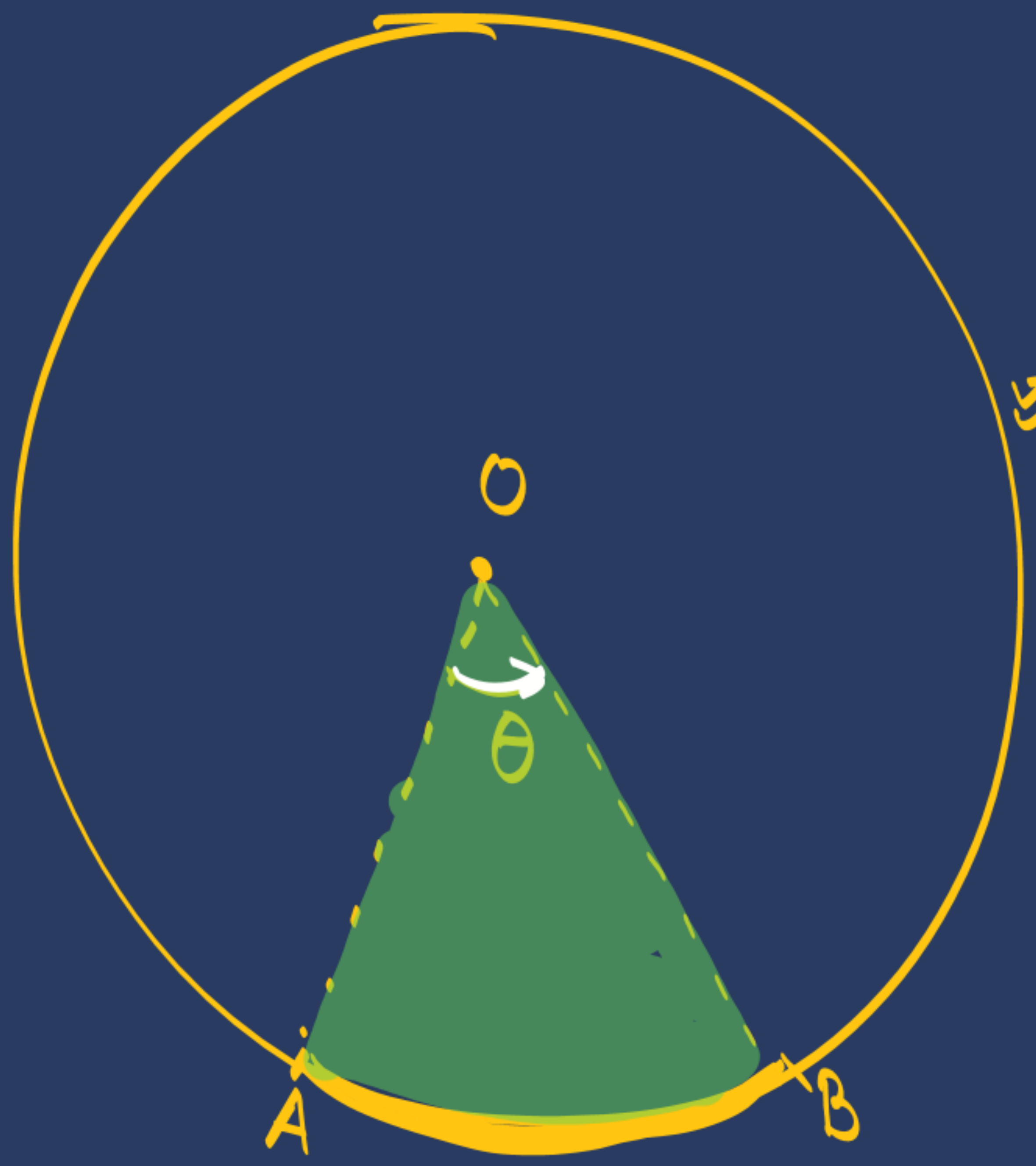
Arc : (टुकड़ा)



Sector (Pizza slice)



Area of Sector :-



$360^\circ \longrightarrow \pi r^2$

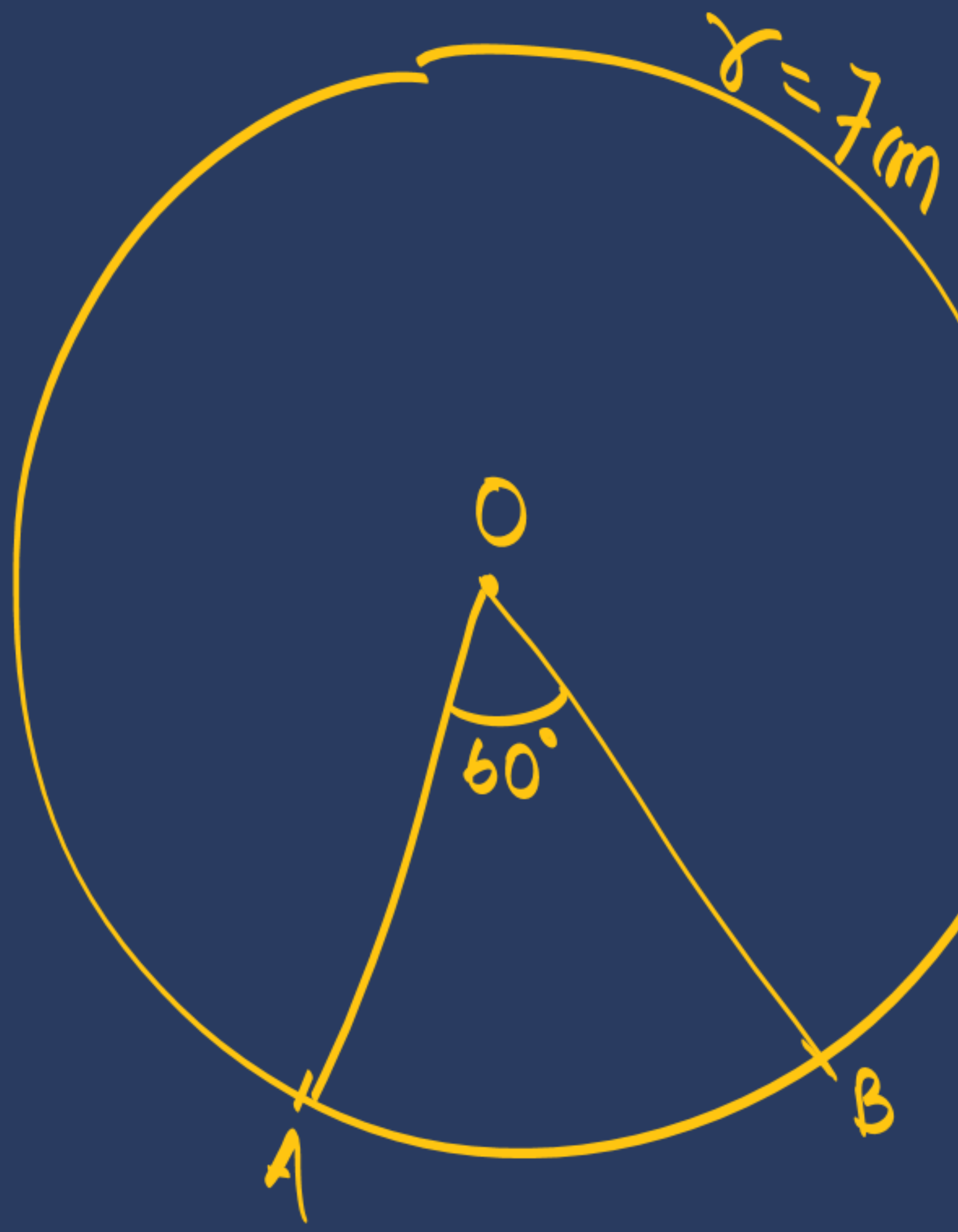
$1^\circ \longrightarrow \dfrac{\pi r^2}{360}$

$\theta^\circ \longrightarrow \theta \times \dfrac{\pi r^2}{360}$

$$\boxed{\text{Area of minor Sector} = \frac{\theta}{360}\pi r^2}$$

Ar. Major sector = Ar. circle − minor sector

$= \left[\pi r^2 - \dfrac{\theta}{360}\pi r^2\right]$



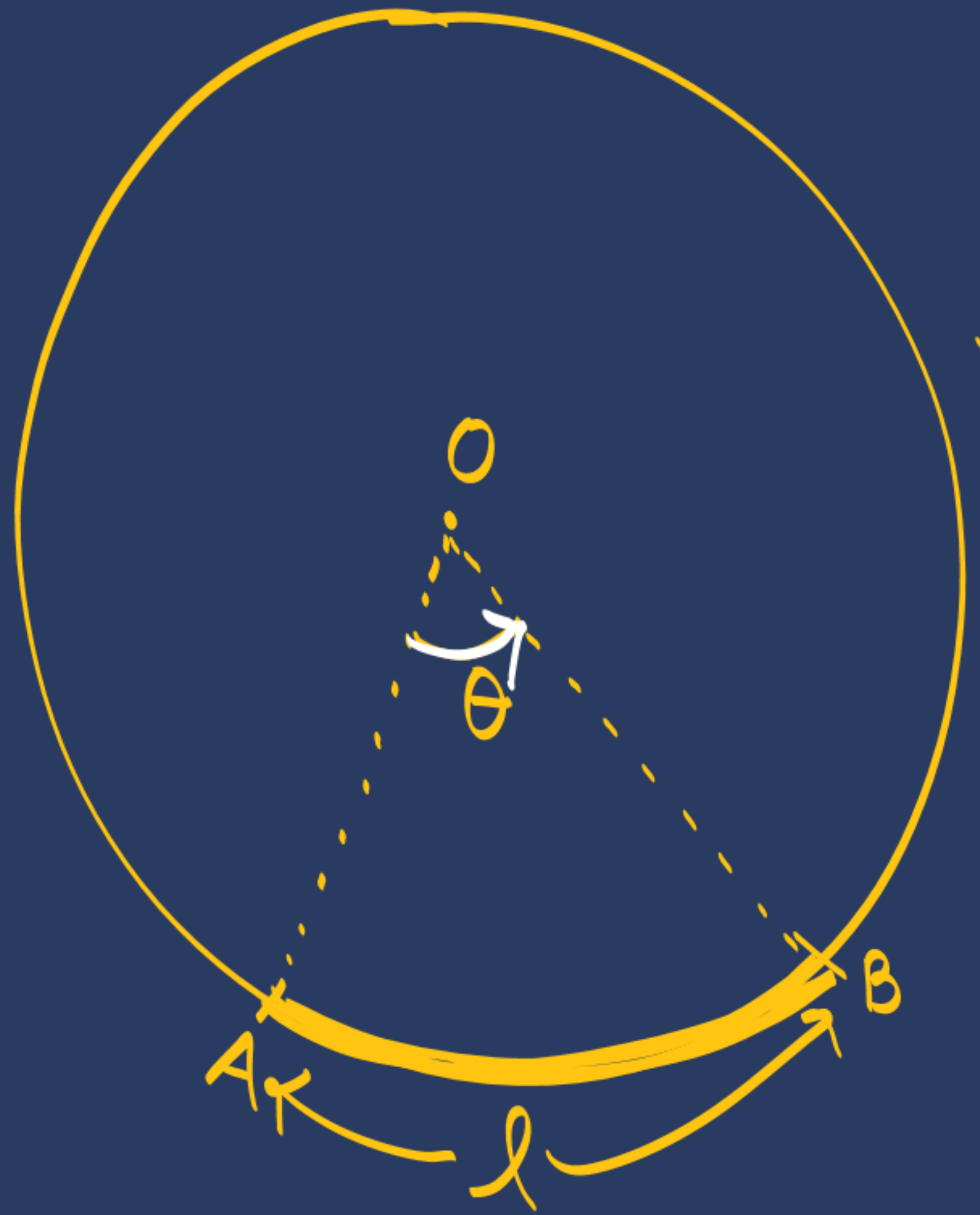
Find area of sector OAB?

$$A = \frac{\theta}{360}\pi r^2$$

$$= \frac{\cancel{60}}{_{3}\cancel{6}\cancel{360}} \times \frac{\cancel{22}^{11}}{\cancel{7}} \times \cancel{7} \times 7$$

$$\boxed{A = \left(\frac{77}{3}\right) cm^2}$$

Length of Arc : (l)



$$360^\circ \longrightarrow 2\pi r$$

$$1^\circ \longrightarrow \frac{2\pi r}{360}$$

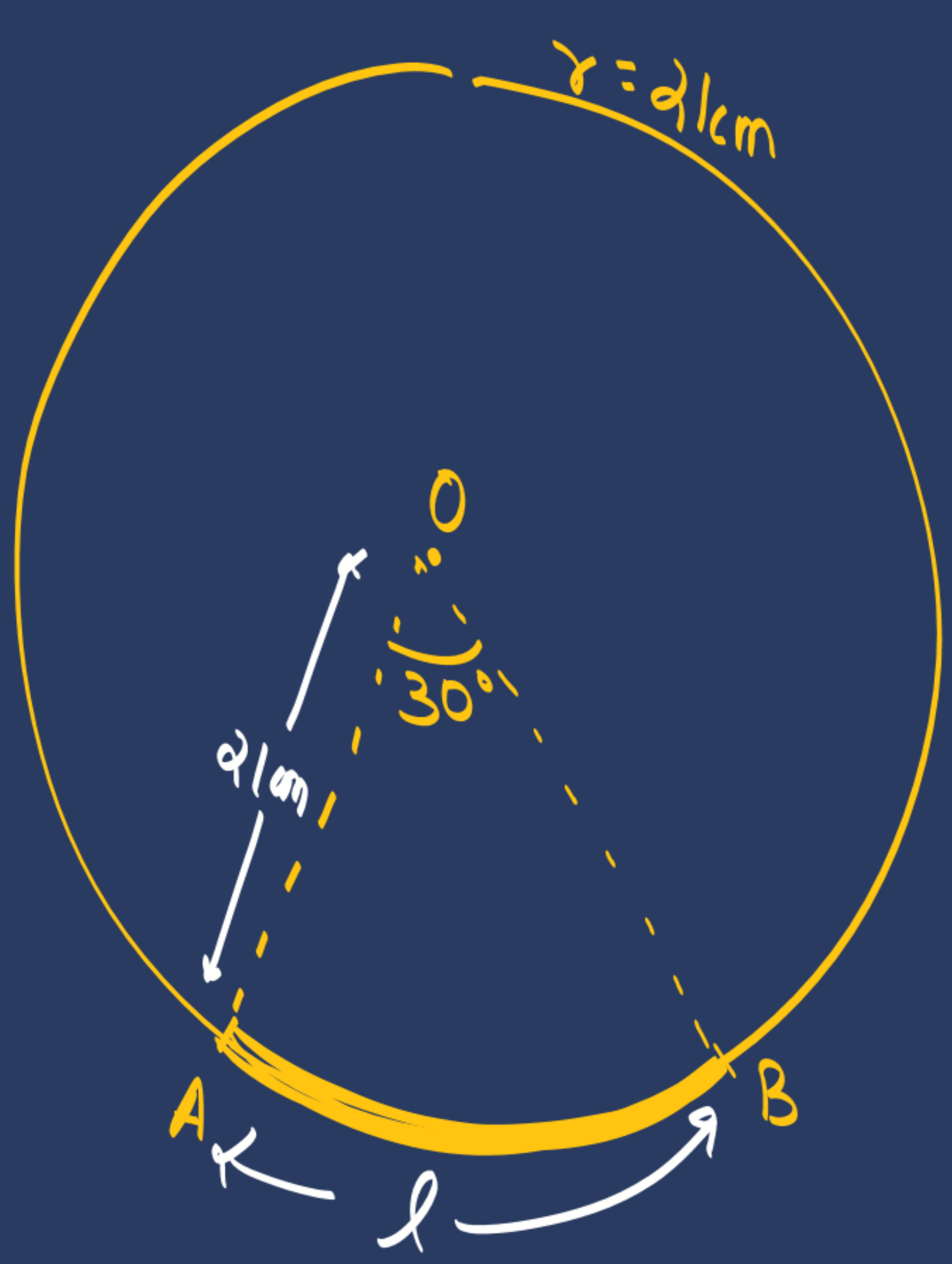
$$\theta^\circ \longrightarrow \theta \times \frac{2\pi r}{360}$$

$$\boxed{\text{Length of minor Arc}(l) = \frac{\theta}{360} \times 2\pi r}$$

Length of major Arc = Circum − Minor Arc

$$= \left[2\pi r - \frac{\theta}{360} 2\pi r\right]$$

#LP:



Length of Arc?

$$l = \frac{\theta}{360} 2\pi r$$

$$= \frac{\cancel{30}}{\cancel{360}} \times \cancel{2} \times \frac{\cancel{22}^{11}}{\cancel{7}} \times \cancel{21}^{3}$$

$$\boxed{l = 11\text{cm}}$$

Length $\longrightarrow$ m, cm, mm, km - - -

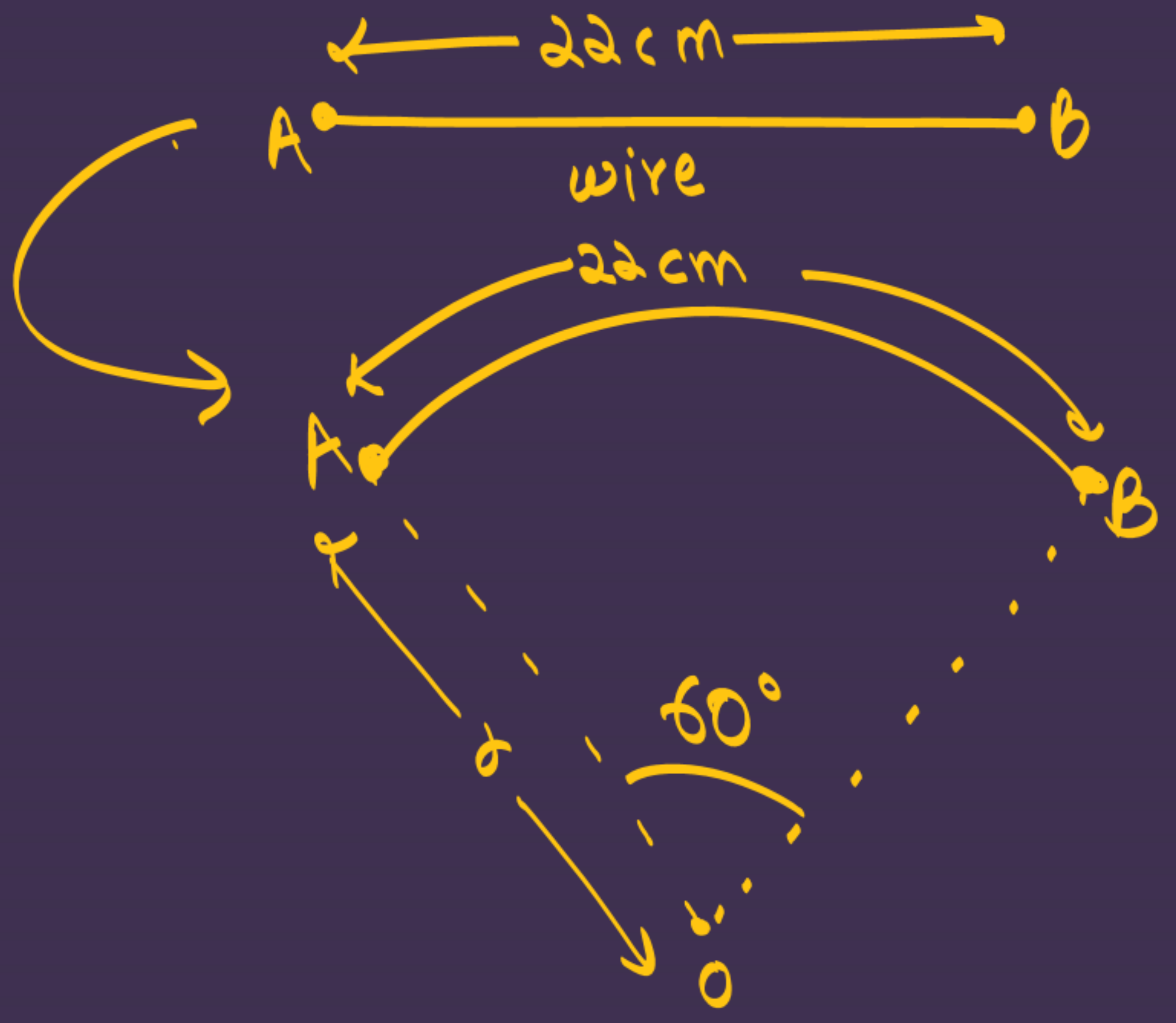
Area $\longrightarrow$ m^2, cm^2, mm^2, km^2 - - -

Volume $\longrightarrow$ m^3, cm^3, mm^3, km^3 - - -

$$\boxed{\begin{array}{l} \text{Area of Sector} \Rightarrow \dfrac{\theta}{360}\pi r^2 \\[2ex] \text{Length of Arc} \Rightarrow \dfrac{\theta}{360} 2\pi r \end{array}}$$

#LP: A piece of wire 22 cm long is bent into the form of an arc of a circle subtending an angle of 60° at its centre. Find the radius of the circle. [Use $\pi = \frac{22}{7}$]

[CBSE 2020]



$l = 22\text{ cm}$

$\Rightarrow \frac{\theta}{360} 2\pi r = 22$

$\Rightarrow \frac{\cancel{60}}{\cancel{360}_{6}} \times \cancel{2} \times \frac{\cancel{22}}{7} \times r = \cancel{22}^{1}$

$\Rightarrow \frac{r}{21} = 1$

$\rightarrow \boxed{r = 21\text{ cm}}$

#LP: A car has two wipers which do not overlap. Each wiper has a blade of length 25cm sweeping through an angle of 115°. Find the total area cleaned at each sweep of the blades

[CBSE 2019,23]

Total area cleaned

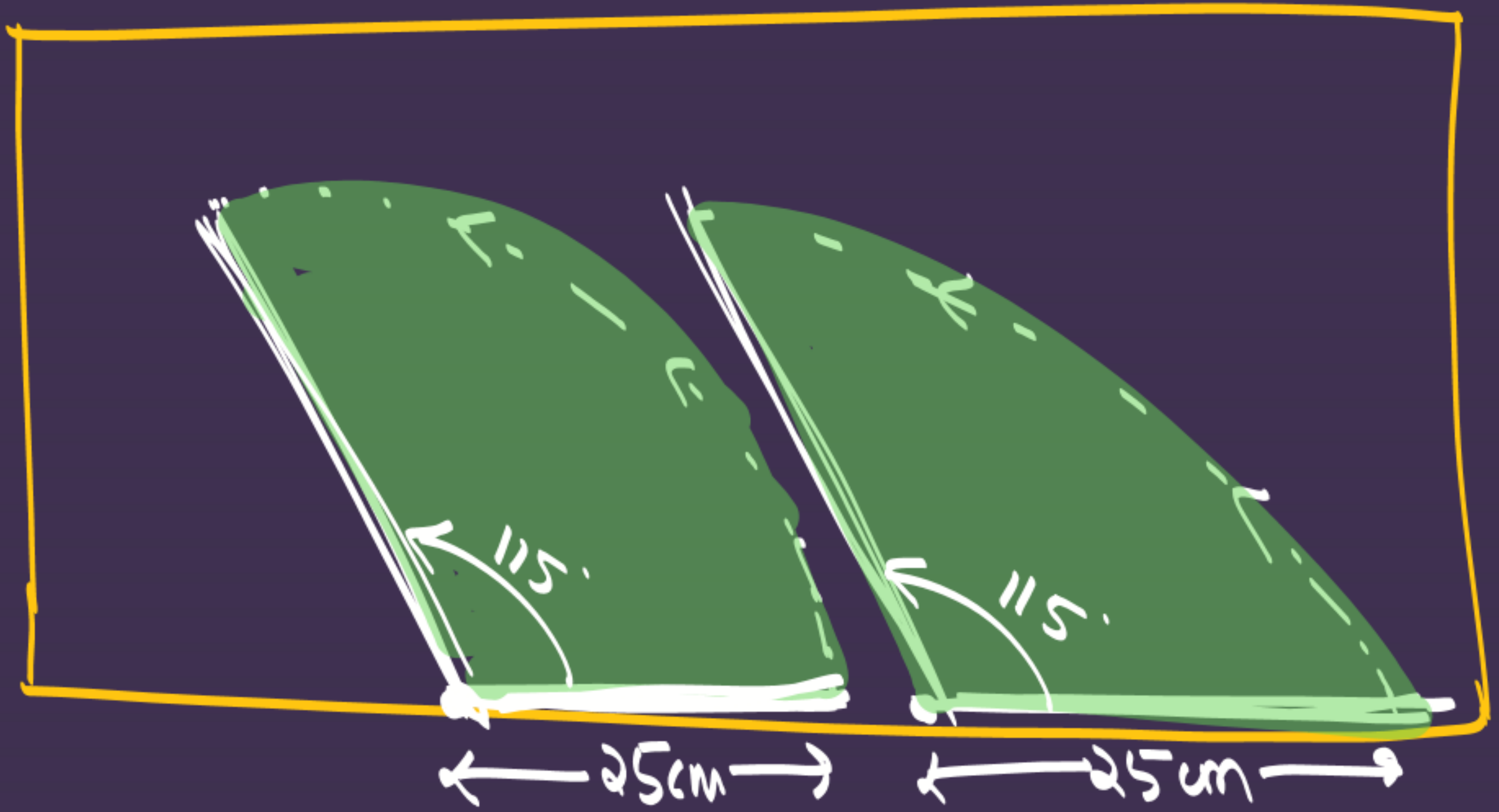
$= 2 \times (\text{Ar. of sector})$

$\Rightarrow 2 \left[\frac{\theta}{360} \pi r^2\right]$

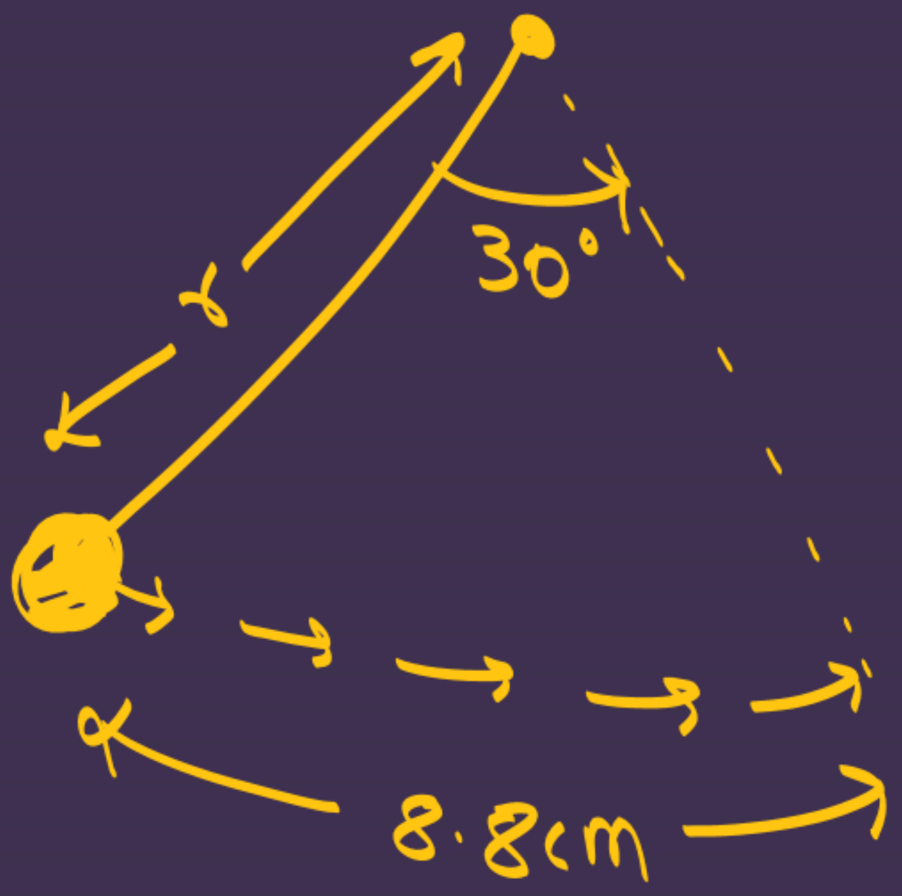
$\Rightarrow 2 \times \left[\frac{115}{360} \times \frac{22}{7} \times 25 \times 25\right] cm^2$

$\downarrow$

$A = 1254.96 \, cm^2$



#LP: A pendulum swing through an angle of 30° and describes an arc 8.8 cm in length. Find the length of the pendulum.



$$l = 8.8\,\text{cm}$$

$$\frac{\theta}{360}\, 2\pi r = 8.8$$

$$\Rightarrow \frac{\cancel{30}}{\cancel{360}} \times \cancel{2} \times \frac{\cancel{22}}{7} \times r = \frac{\cancel{88}}{10}$$

$$\frac{r}{21} = \frac{8}{10}$$

$$r = \frac{21 \times 8}{10}$$

length of pendulum → $\boxed{r = 16.8\,\text{cm}}$

#LP: An umbrella has 8 ribs which are equally spaced. Assuming umbrella to be a flat circle of radius 45 cm, find the area between the two consecutive ribs of the umbrella.

एक के बाद एक

$$A = \frac{\theta}{360}\pi r^2$$

$$A = \left[\frac{45}{360} \times \frac{22}{7} \times 45 \times 45\right] cm^2$$

✓

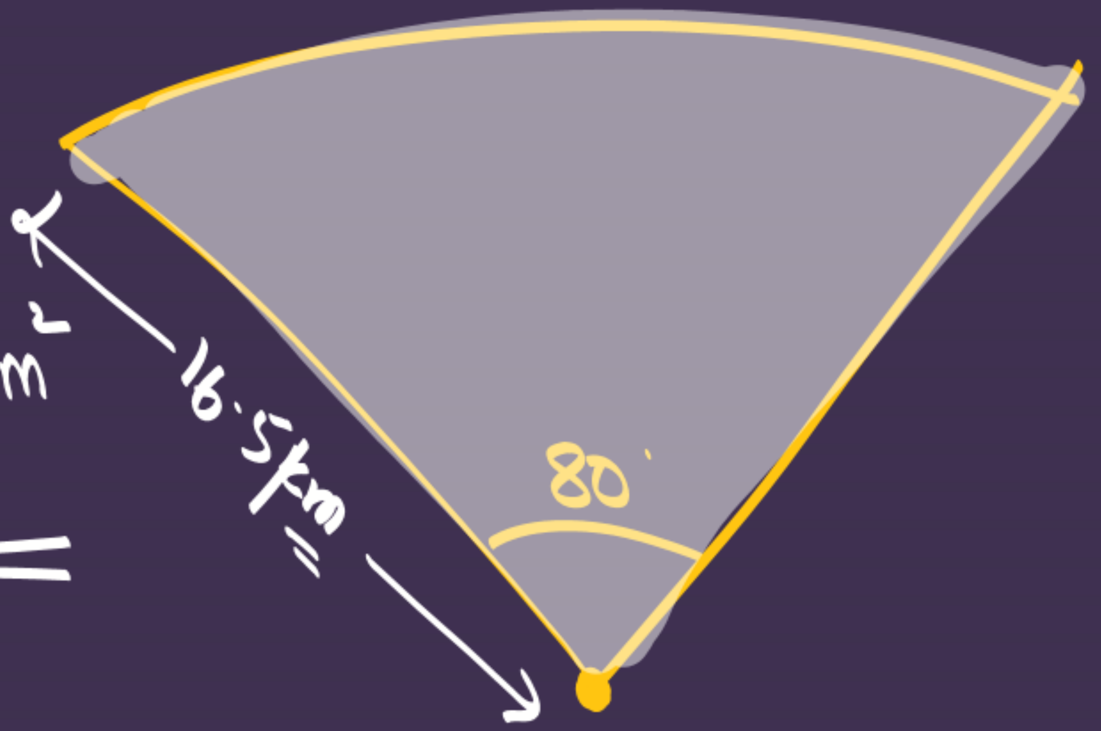
$$\theta = \frac{\cancel{360}^{45}}{\cancel{8}}$$

$$\boxed{\theta = 45°}$$

(Top view)

#LP: To warn ships for underwater rocks, a lighthouse spreads a red coloured light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which the ships are warned. (Use π = 3.14)

top view



$$\text{Ar. sector} = \frac{\theta}{360} \times \pi r^2$$

$$A \Rightarrow \left[\frac{80}{360} \times 3.14 \times 16.5 \times 16.5\right] km^2$$

#LP: A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in Fig. Find :

(i) the total length of the silver wire required.

(ii) the area of each sector of the brooch.

$d = 35\,mm$

$r = \frac{35}{2}\,mm$

(i) total length $\Rightarrow 2\pi r + 5d$

$\Rightarrow \left[\cancel{2} \times \frac{22}{\cancel{7}} \times \frac{\cancel{35}^{5}}{\cancel{2}}\right] mm + (5 \times 35\,mm)$

$\Rightarrow 110\,mm + 175\,mm$

$\boxed{\text{total length} \Rightarrow 285\,mm}$

$\theta = \frac{36\cancel{0}}{1\cancel{0}}$

$\boxed{\theta = 36^\circ}$

(ii) Ar. sector $\Rightarrow \frac{\theta}{360}\pi r^2 \Rightarrow \left(\frac{36}{360} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2}\right) mm^2$

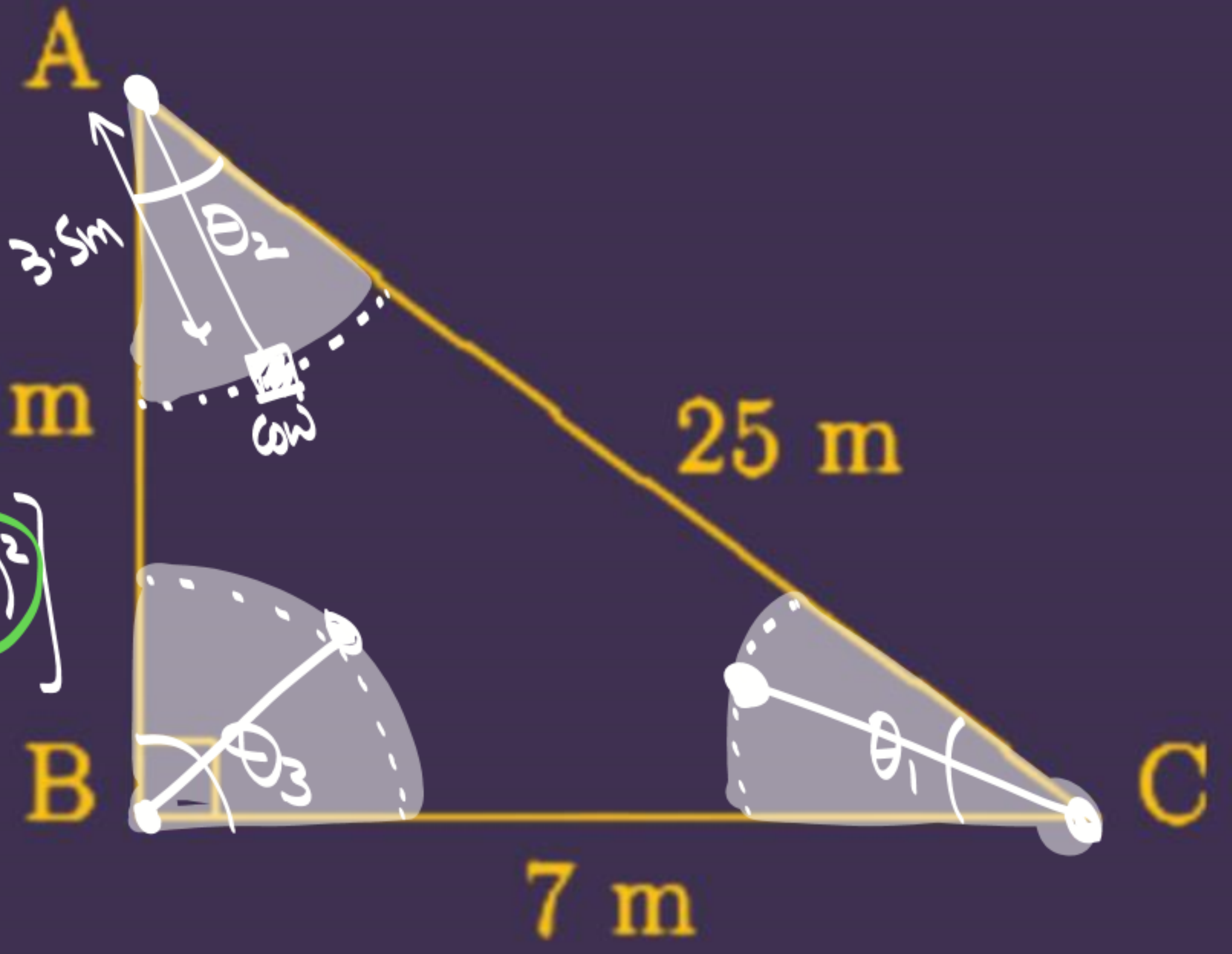
#LP: Sides of a right triangular field are 25m, 24m and 7m. At the three corners of the field, a cow, a buffalo and a horse are tied separately with ropes of 3.5 m each to graze in the field. Find the area of the field that cannot be grazed by these animals.

Area cannot be grazed $\Rightarrow [\Delta \text{ area}] - [\text{ar of } 3 \text{ sectors}]$

$$\Rightarrow \left[\frac{1}{\cancel{2}} \times 7 \times \cancel{24}^{12}\right] - \left[\frac{\theta_1}{360}\pi(3.5)^2 + \frac{\theta_2}{360}\pi(3.5)^2 + \frac{\theta_3}{360}\pi(3.5)^2\right]$$

$$\Rightarrow [7 \times 12] - \frac{\pi(3.5)^2}{360}(\theta_1 + \theta_2 + \theta_3)$$

$$\Rightarrow \left[84 - \frac{22}{7} \times \frac{(3.5)^2}{360} \times 180\right] m^2$$



#LP: . A horse is tied to a peg at one corner of a square shaped Fig.) Field of side 15 m by means of a 5 m long rope (see Fig.). Find (i) the area of that part of the field in which the horse car graze. (ii) the increase in the grazing area if the rope were 10 long instead of 5 m . (Use $\pi = 3.14$)

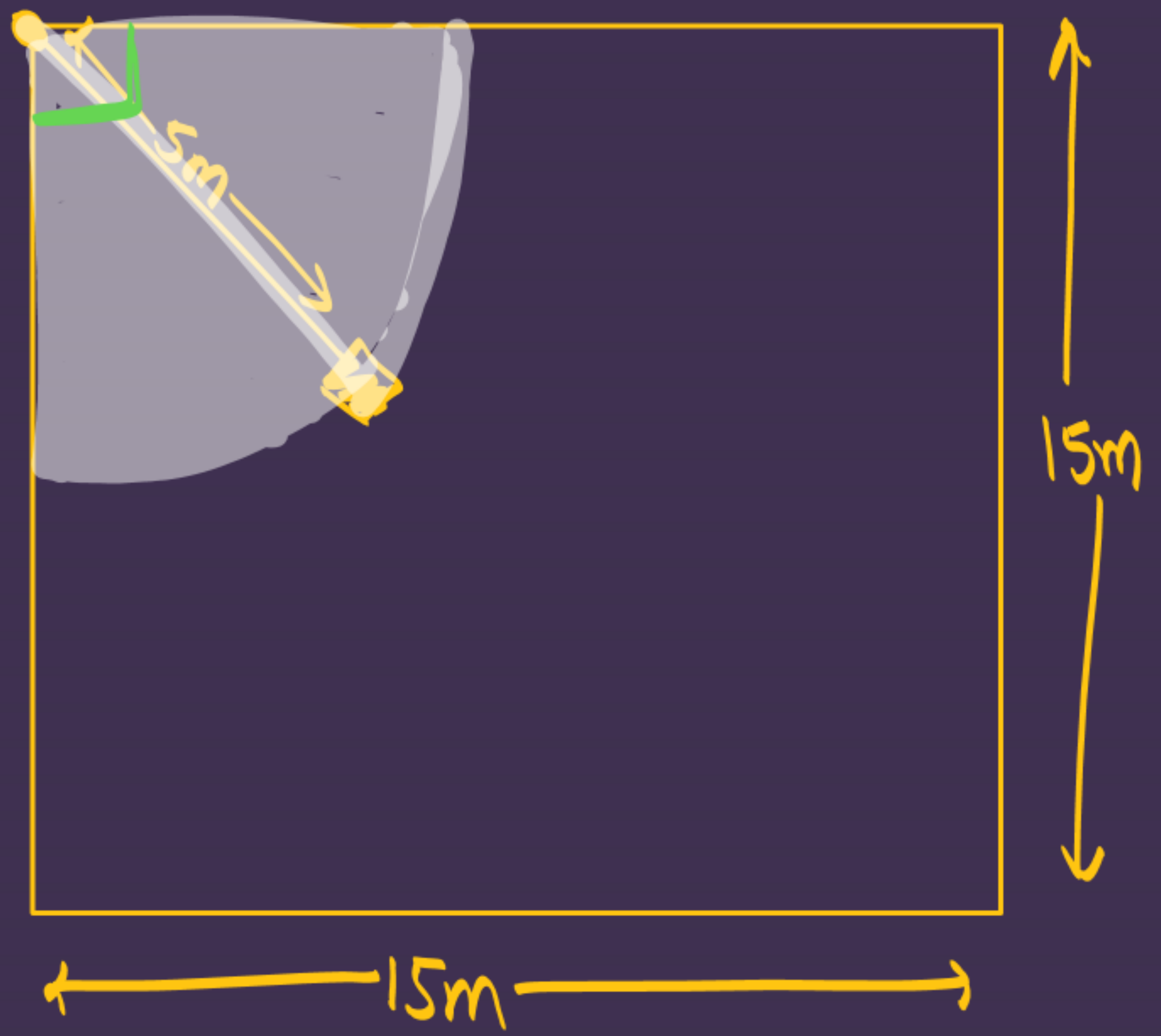
[CBSE 2021]

(i) Ar. sector $\Rightarrow \frac{\theta}{360} \pi r^2$

$$A_1 \rightarrow \left[\frac{90}{360} \times 3.14 \times 5 \times 5\right] m^2$$

(ii) 10m rope $A_2 = \left[\frac{90}{360} \times 3.14 \times 10 \times 10\right] m^2$

Increase $\Rightarrow (A_2 - A_1)$



#LP: The length of the minute hand of a clock is 14cm. find the area swept by minute hand in 10 minutes

[CBSE 2012]

Area swept in 10min $\Rightarrow \frac{\theta}{360} \pi r^2$

$\Rightarrow \left[\frac{60}{360} \times \frac{22}{7} \times 14 \times 14\right] cm^2$

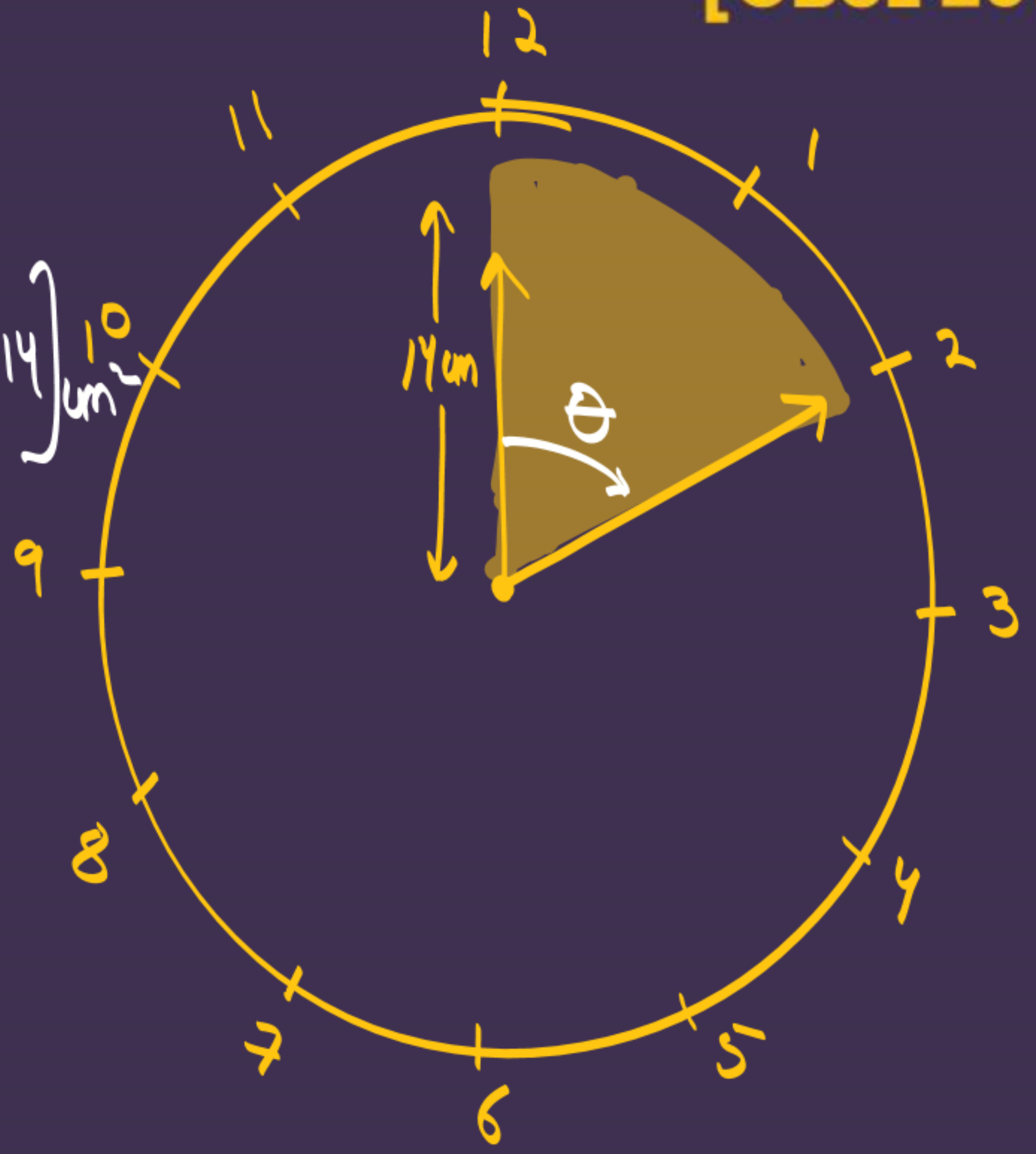
Minute hand

$60 \text{ min} \longrightarrow 360^\circ$

$1 \text{ min} \longrightarrow \left(\frac{360}{60}\right)^\circ$

$10 \text{ min} \longrightarrow \left(10 \times \frac{360}{60}\right)^\circ$

$\boxed{\theta \Rightarrow 60^\circ}$



Second's hand

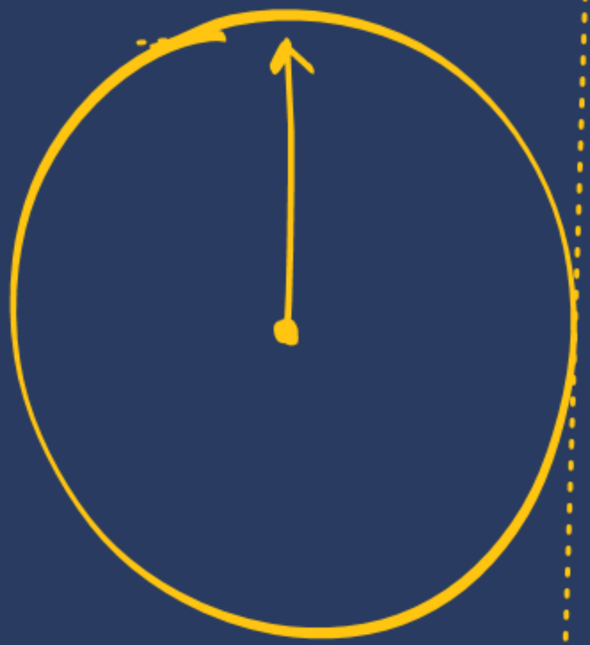
x sec $\rightarrow \theta = ??$

60 sec $\rightarrow (360^\circ)$

1 sec $\rightarrow \left(\frac{360}{60}\right)^\circ$

x sec $\rightarrow \left(x \times \frac{360}{60}\right)^\circ$

θ



Hour hand :-

x hr $\rightarrow \theta = ??$

12hr $\rightarrow 360^\circ$

1hr $\rightarrow \left(\frac{360}{12}\right)^\circ$

x hr $\rightarrow \left(x \times \frac{360}{12}\right)^\circ$

θ

#LP: The minute hand of a clock is 10 cm long. Find the area of the face of the clock described by the minute hand between 9 A.M. and 9.35 A.M.

$$A = \frac{\theta}{360} \pi r^2$$

$$A = \left[\frac{210}{360} \times \frac{22}{7} \times 10 \times 10\right] cm^2$$

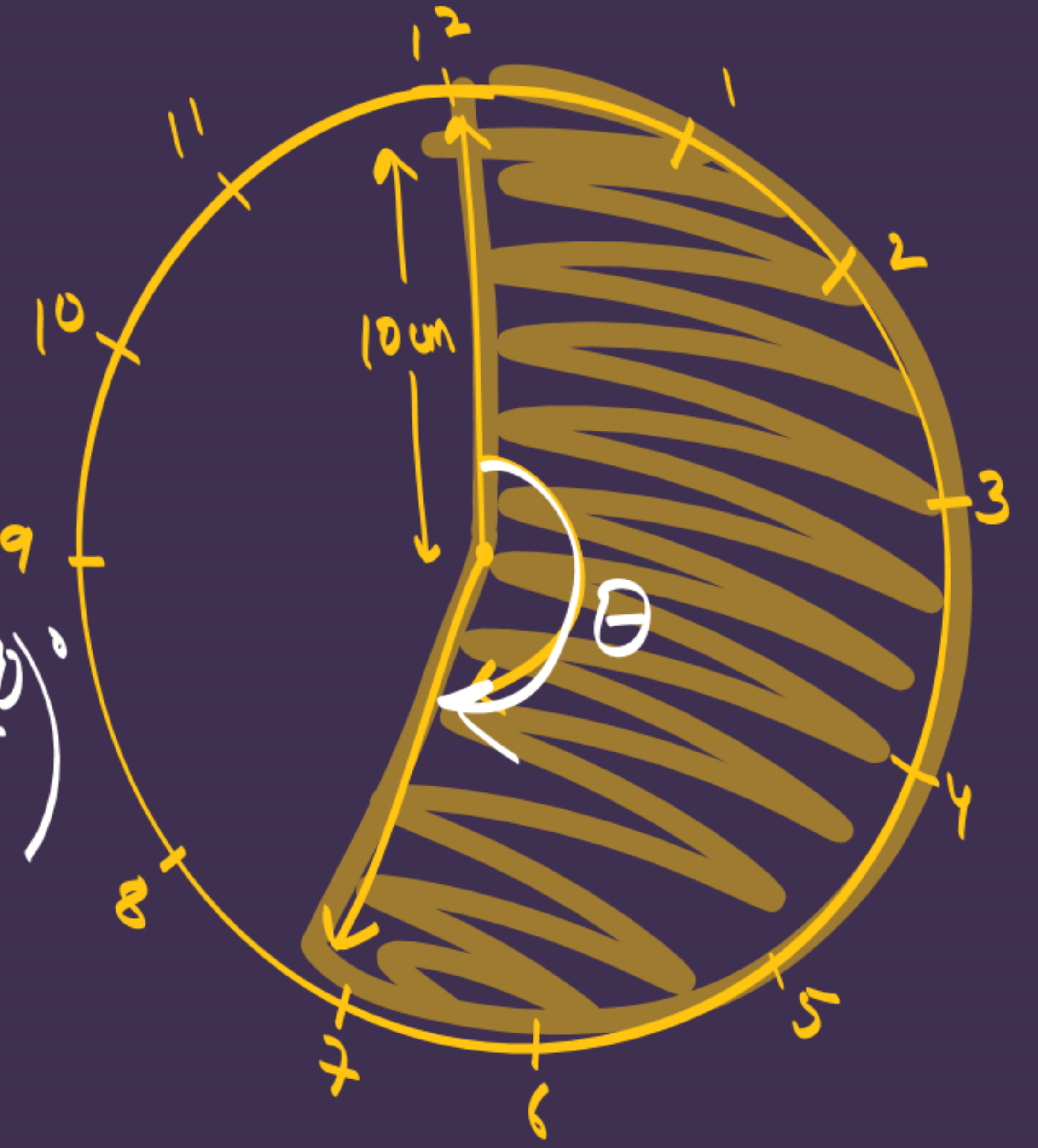
Min hand.

$60 \text{ min} \rightarrow 360^\circ$

$1 \text{ min} \rightarrow \left(\frac{360}{60}\right)^\circ$

$35 \text{ min} \rightarrow \left(35 \times \frac{\cancel{360}^{6}}{\cancel{60}}\right)^\circ$

$$\boxed{\theta = 210^\circ}$$



#LP: The minute-hand of a circle is 84 cm long. The distance covered by the tip of the minute hand from 10:10 A.M. to 10:25 A.M. is :

(a) 44 cm

(b) 88 cm

(c) 132 cm

(d) 176 cm. (CBSE 2022)

Arc. length ⇒ $\frac{\theta}{360} 2\pi r$

$\Rightarrow \left(\frac{\cancel{90}}{\cancel{360}_{4}} \times 2 \times \frac{22}{\cancel{7}} \times \cancel{84}^{\,12\;6}\right) cm$

$\boxed{l \Rightarrow 132 cm}$

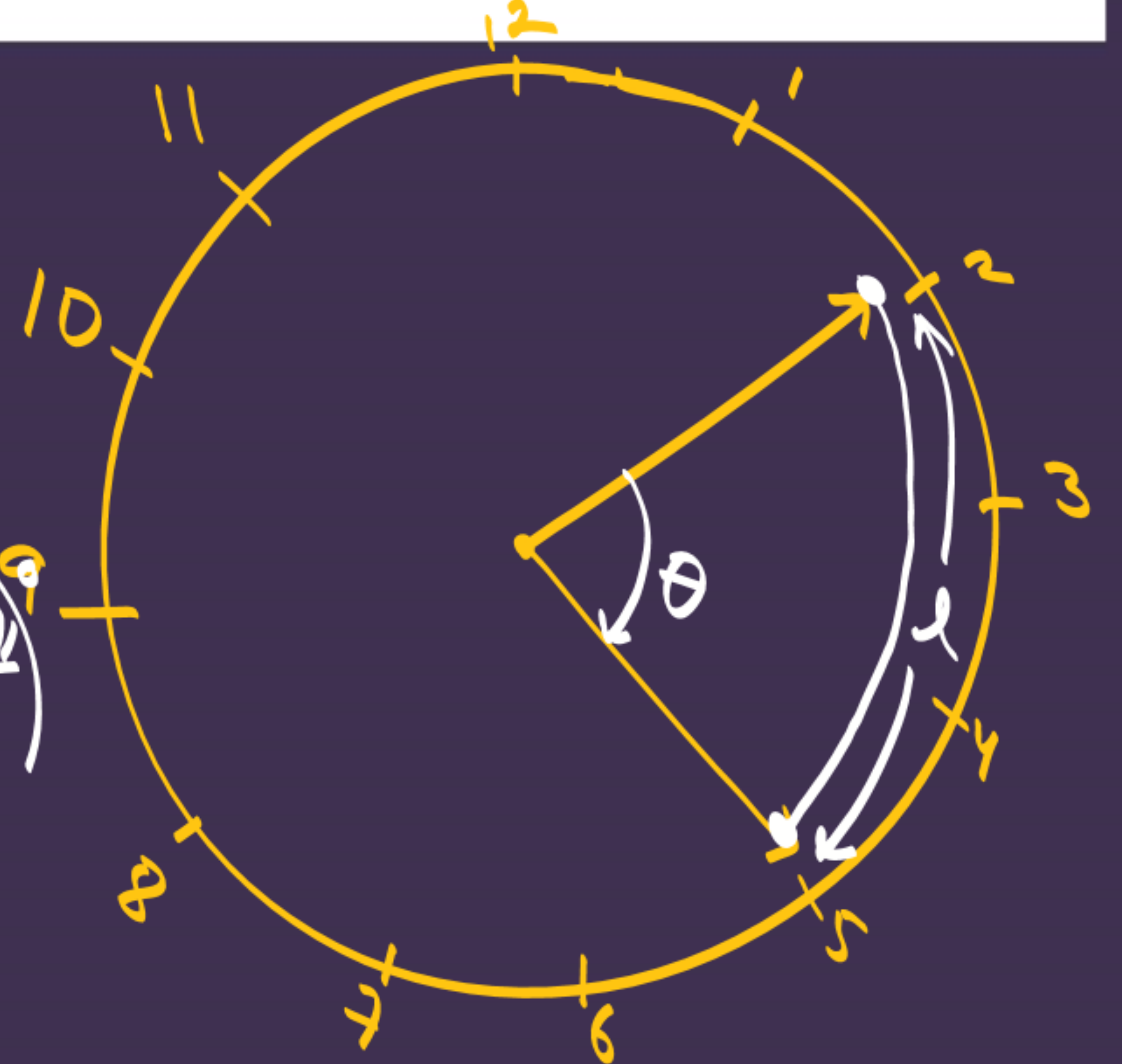
Min. hand

60 min → 360°

1 min → $\left(\frac{360}{60}\right)^\circ$

15 min → $\left(15 \times \frac{\cancel{360}^{\,6}}{\cancel{60}}\right)^\circ$

$\boxed{\theta \Rightarrow 90^\circ}$



#LP: Figure shows a sector of a circle, centre O, containing an angle θ. .
Prove that
(i) Perimeter of the shaded region is $r\left(\tan\theta+\sec\theta+\frac{\pi\theta}{180}-1\right)$

(i) Perimeter shaded $\Rightarrow \left[AB + \overset{\frown}{AC} + BC\right]$

AB

$\cos\theta = \frac{r}{OB}$

$\boxed{OB = \frac{r}{\cos\theta}}$

$\tan\theta = \frac{AB}{r}$

$\boxed{AB = r\tan\theta}$

AC

$\boxed{l = \frac{\theta}{360} 2\pi r}$

BC

$BC = OB - r$

$BC = \frac{r}{\cos\theta} - r$

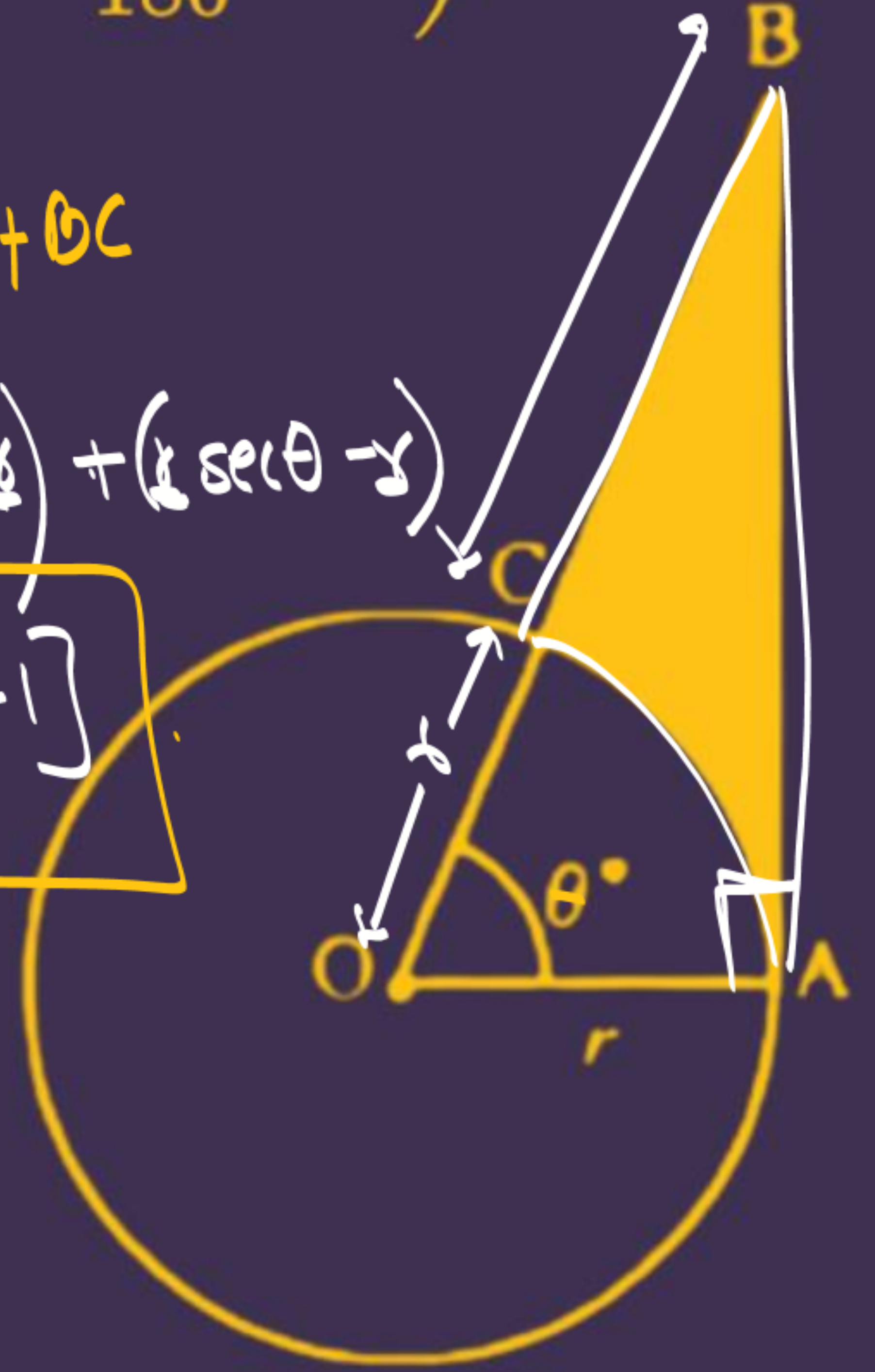
$\boxed{BC = r\sec\theta - r}$

Final Ans $\Rightarrow P = AB + \overset{\frown}{AC} + BC$

$P \Rightarrow (r\tan\theta) + \left(\frac{\theta}{\cancel{360}_{180}} \cancel{2}\pi r\right) + (r\sec\theta - r)$

$\boxed{P = r\left[\tan\theta + \frac{\pi\theta}{180} + \sec\theta - 1\right]}$

HP



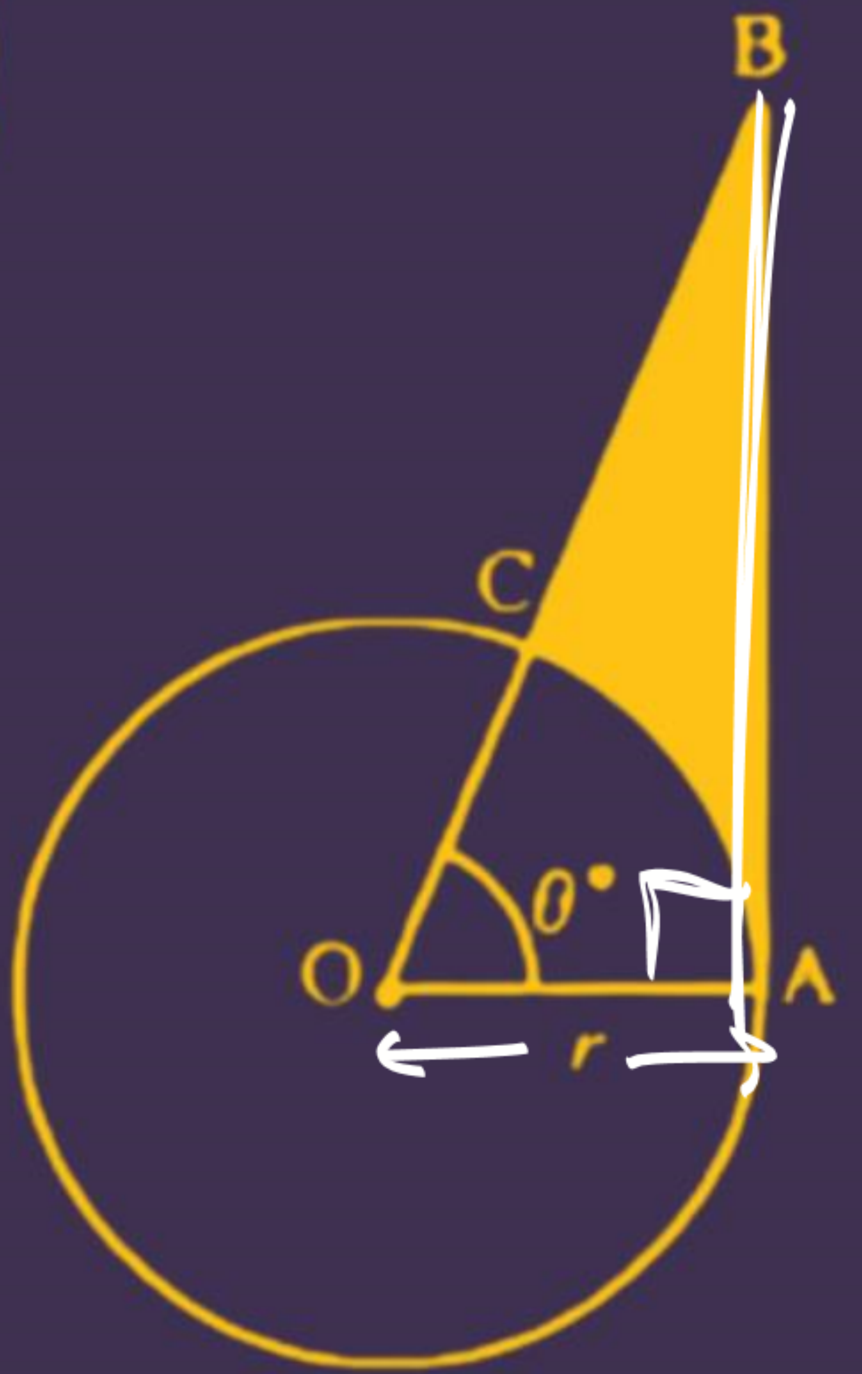
(ii) Area of shaded region is $\dfrac{r^2}{2}\left(\tan\theta - \pi\dfrac{\theta}{180}\right)$

Ar. shaded = Δ area − sector area

$$\Rightarrow \left[\frac{1}{2}\times b\times h\right] - \left[\frac{\theta}{360}\pi r^2\right]$$

$$\Rightarrow \left[\frac{1}{2}\times r\times r\tan\theta\right] - \left[\frac{\theta}{\cancel{360}_{180}}\pi r^2\right]\times\frac{\cancel{2}}{2}$$

$$\Rightarrow \frac{r^2}{2}\left[\tan\theta - \frac{\pi\theta}{180}\right]$$

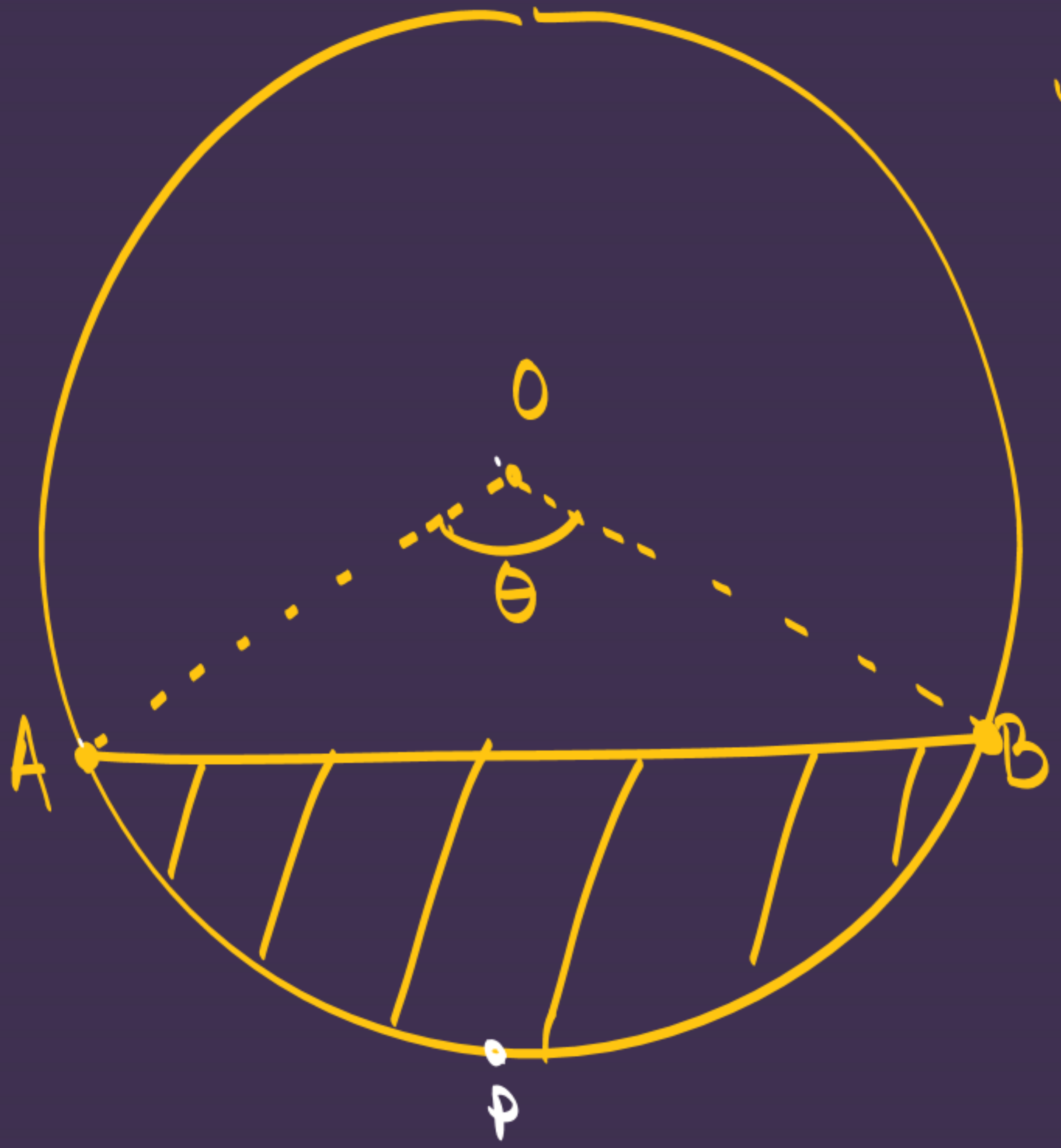


Ar. sector $= \frac{\theta}{360} \pi r^2$

Arc. length$(l) = \frac{\theta}{360} 2\pi r$

$\boxed{\text{Segment} = ??}$

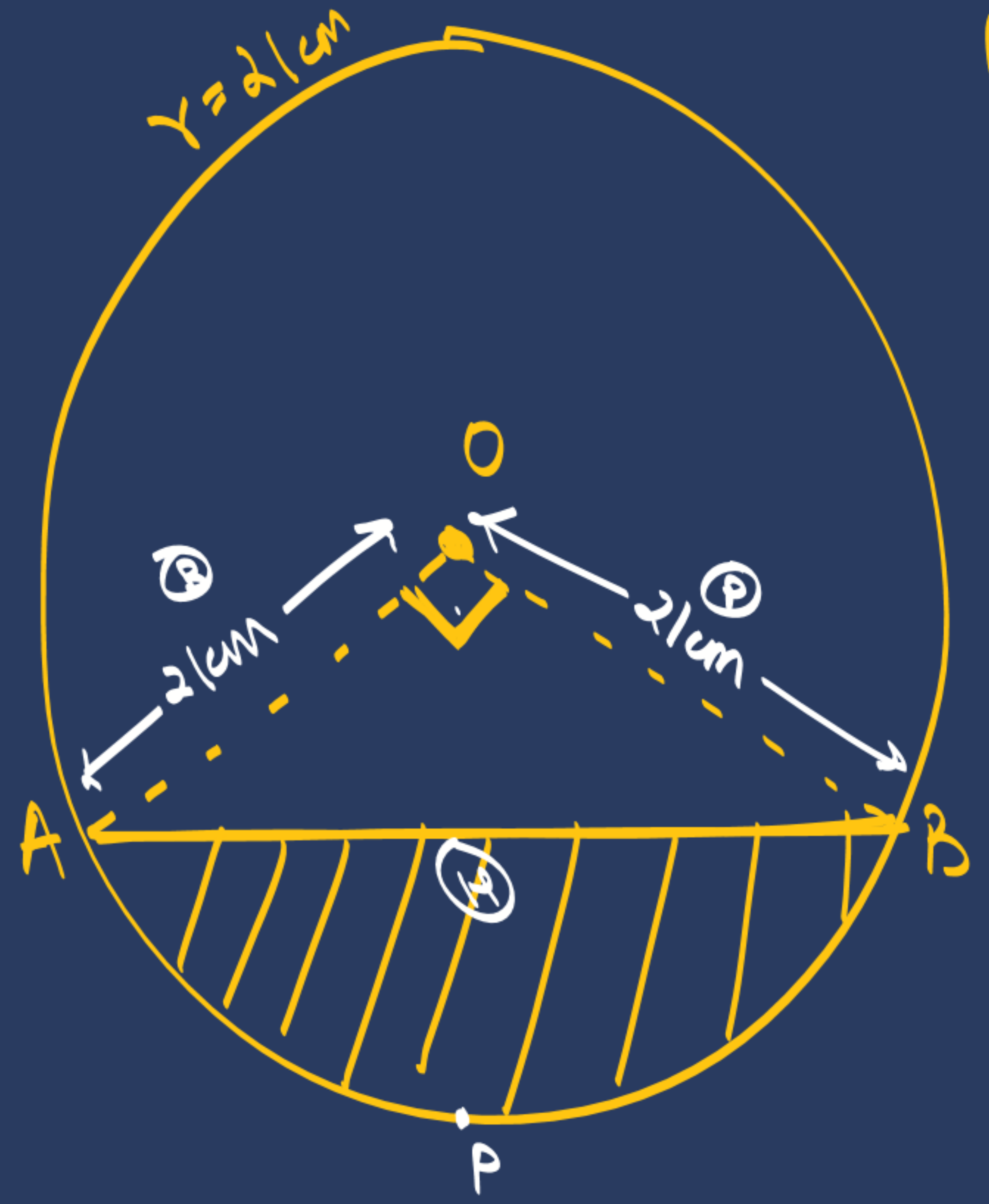
#Area of Segment: $\Rightarrow$ [Area of Sector − Ar. Δ]



$$\boxed{\text{Ar. minor Segment} \Rightarrow [\text{Ar. sector } OAPB - \Delta OAB]}$$

Ar. major segment $\Rightarrow$ (ar Circle − ar. minor seg.)

1P:-



Find Area of segment ?

Ar segment (APB) = Ar. sector (OAPB) - ar ΔOAB

$$\Rightarrow \left[\frac{90}{360} \times \frac{22}{7} \times 21 \times 21\right] - \left[\frac{1}{2} \times 21 \times 21\right] cm^2$$

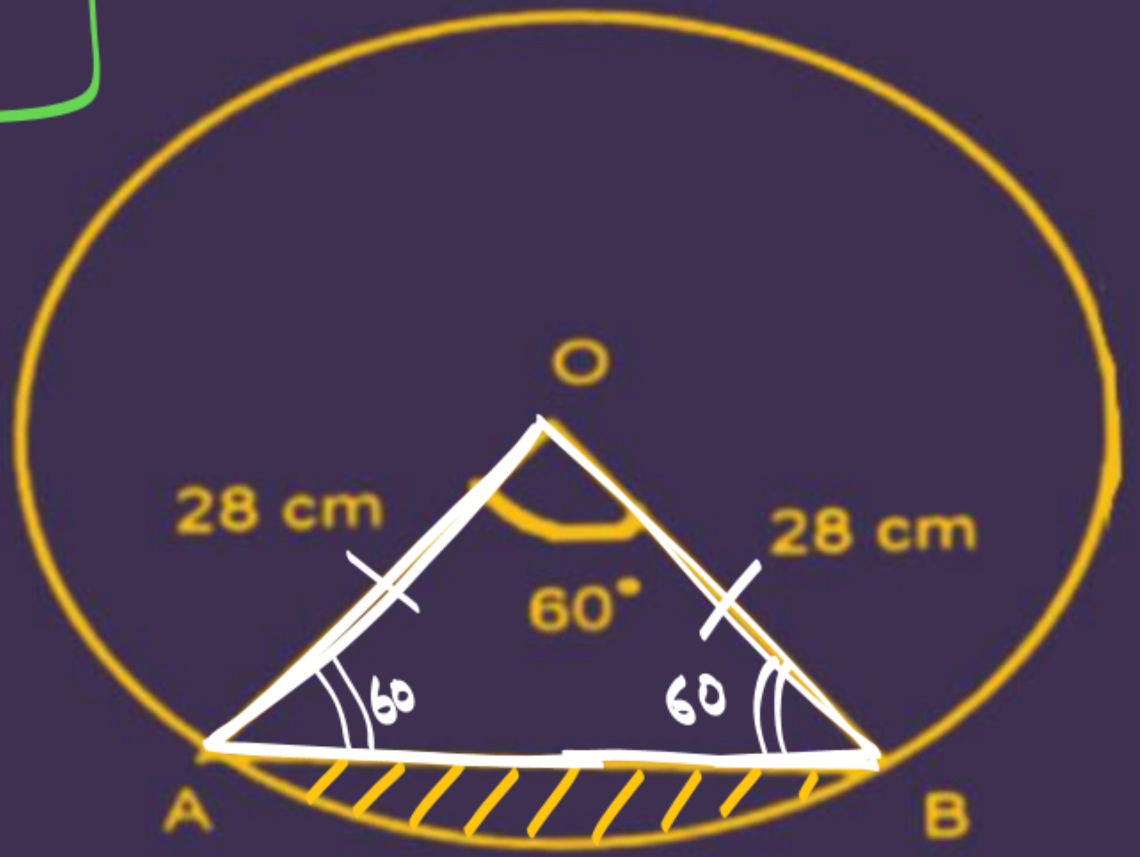
90° → Ar. $\Delta = \frac{1}{2} \times b \times h$

60°

#LP: A chord of a circle of radius 28 cm subtends an angle of 60° at the centre of the circle. The area of the minor segment is: ($\sqrt{3}$=1.732)

(A) 60.256 cm^2
(B) 339.47 cm^2
(C) 410.67 cm^2
(D) 71.20 cm^2

$$\boxed{\text{Ar. Eq.}\Delta = \frac{\sqrt{3}}{4}(\text{side})^2}$$



ASP ⇒ $60 + 2x = 180$

$\cancel{2}x = \cancel{120}\ 60$

$\boxed{x = 60^\circ}$

Ar. segment ⇒ Ar sector − Ar. Δ(OAB)

$$\Rightarrow \left(\frac{60}{360} \times \frac{22}{7} \times 28 \times 28\right) - \left(\frac{\sqrt{3}}{4}(28)^2\right) cm^2$$

↓

~~FLP~~ 120°

LP: Find the area of the segment of circle, given that the angle of the sector 120° and the radious of the circle is 21cm .(Take π = 22/7)

A. segment

$\Rightarrow$ sector $-$ ΔOAB

$\Rightarrow \left[\frac{120}{360} \times \frac{22}{7} \times 21 \times 21\right] - \left(\frac{1}{2} \times AB \times OP\right)$

$(21\sqrt{3})$ $\left(\frac{21}{2}\right)$

$\Rightarrow [22 \times 21] - \left(\frac{1}{2} \times 21\sqrt{3} \times \frac{21}{2}\right) cm^2$

$\boxed{ASP} \rightarrow 120 + 2x = 180$

$2x = 60 \Rightarrow \boxed{x = 30°}$

Draw $OP \perp AB \Rightarrow \boxed{AP = PB}$

$\sin 30 = \frac{OP}{21}$

$\frac{1}{2} = \frac{OP}{21}$

$\boxed{OP = \frac{21}{2} cm}$

$\cos 30 = \frac{AP}{21}$

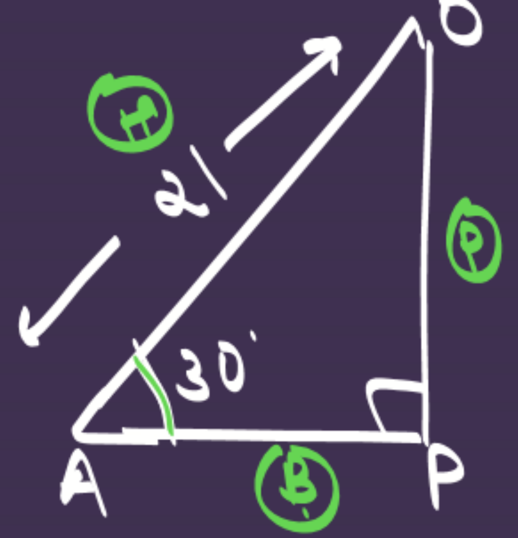
$\frac{\sqrt{3}}{2} = \frac{AP}{21}$

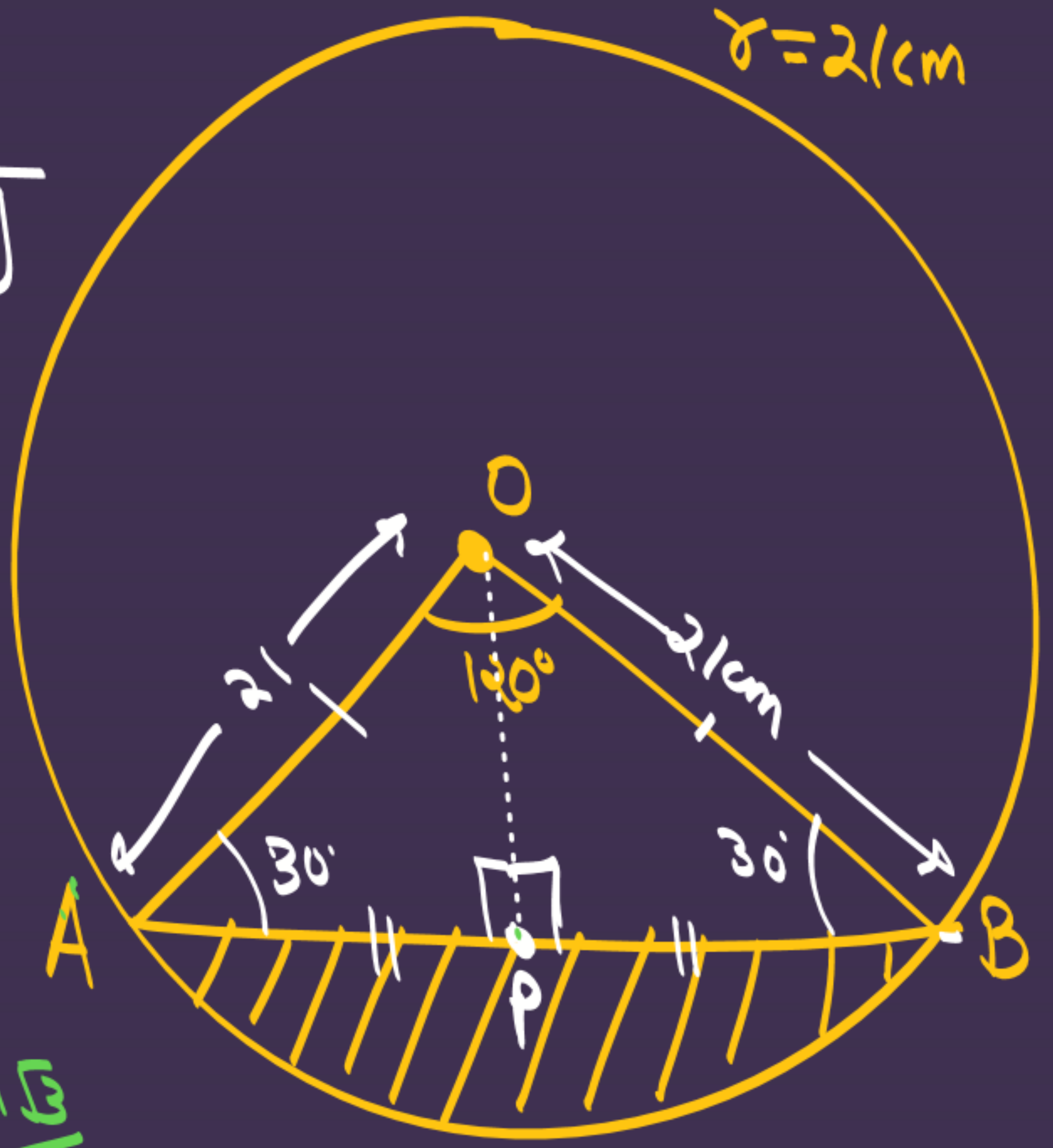
$\boxed{AP = \frac{21\sqrt{3}}{2}}$

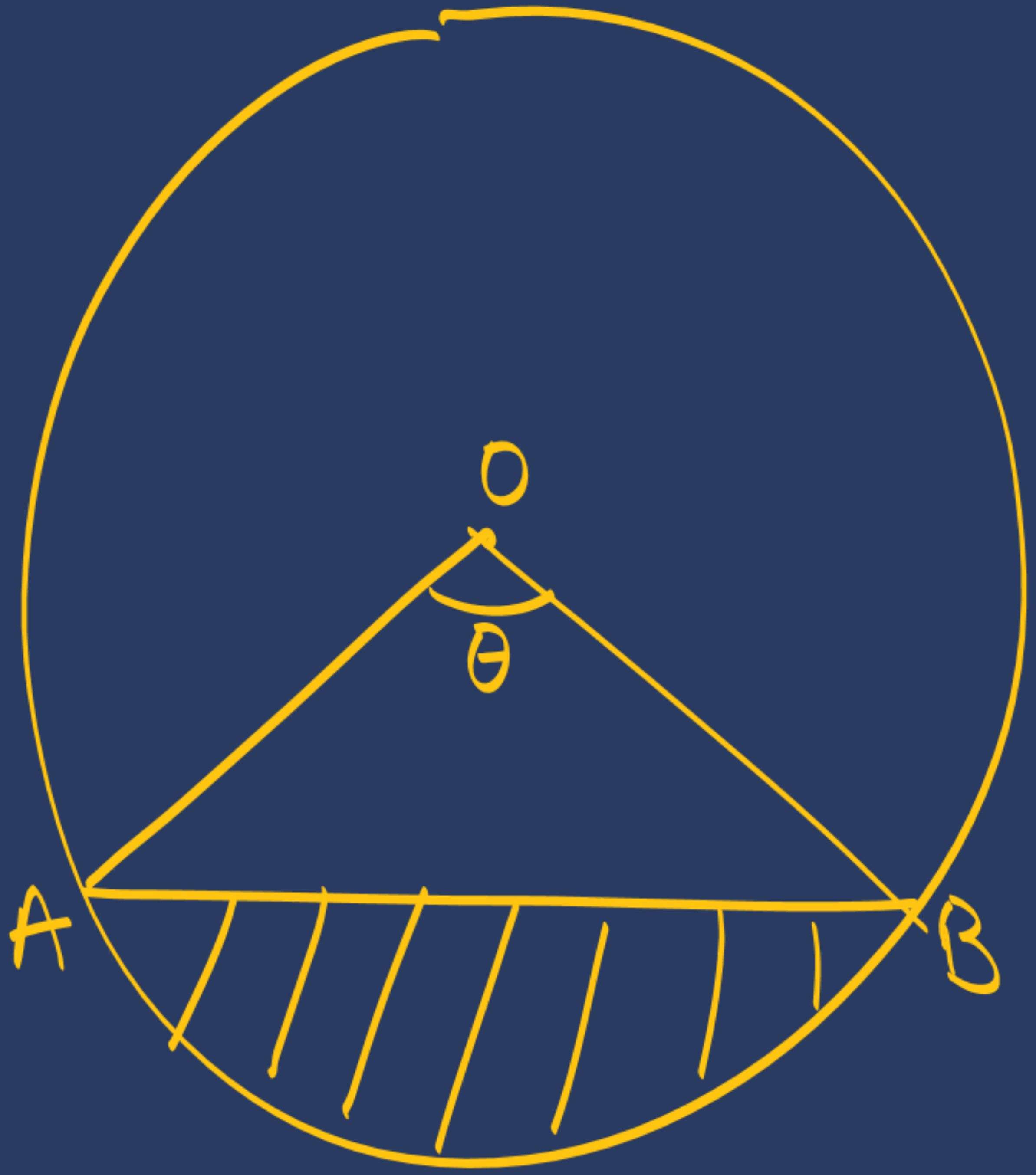
$\rightarrow AB = 2AP$

$= 2 \times \frac{21\sqrt{3}}{2}$

$AB = 21\sqrt{3}$







$\theta = 90 \rightarrow A = \frac{1}{2} \times b \times h$

$\theta = 60^\circ \rightarrow$ eq. $\Delta \Rightarrow A = \frac{\sqrt{3}}{4}(\text{side})^2$

$\theta = 120^\circ \rightarrow A = \frac{1}{2} \times b \times h$

(trigonometry use)

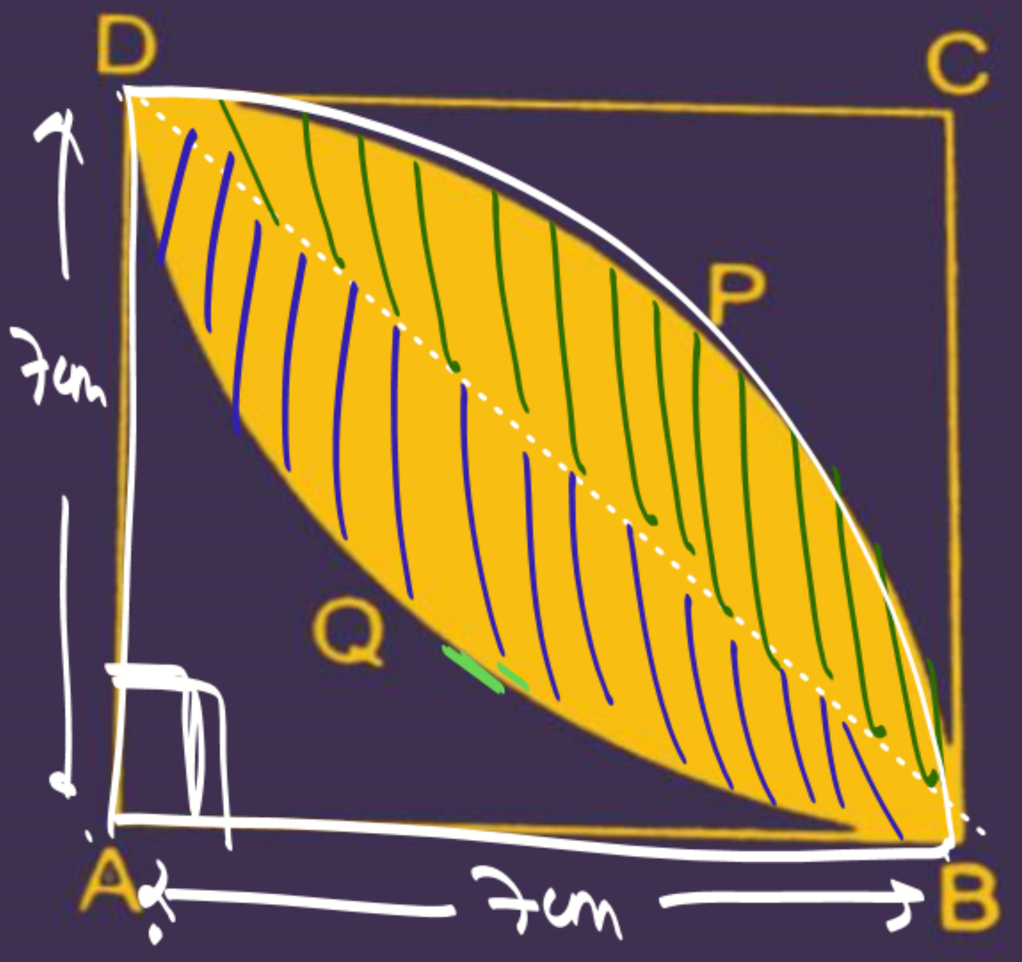
#LP: In the given figure, ABCD is a square of side 7cm, DPBA and DQBC are quadrants of circles each of the radius 7 cm. Find the area of shaded region.

$\boxed{\text{Join BD}}$

Ar. of shaded region $\Rightarrow 2 \times (\text{Ar. segment})$

$\Rightarrow 2\left[\text{Ar. sector} - \text{Ar } \Delta ABD\right]$

$\Rightarrow 2\left[\left(\frac{90}{360} \times \frac{22}{7} \times 7 \times 7\right) - \frac{1}{2} \times 7 \times 7\right]$



#LP: Directions: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(C) Assertion (A) is true but reason (R) is false.

(d) Assertion (A) is false but reason (R) is true.

#LP: **Assertion (A):** The length of the minute hand of a clock is 7 cm. Then the area swept by the minute hand in 5 minute is 77/6 cm^2. T

Reason (R): The length of an arc of a sector of angle q and radius r is given by $l = \frac{\theta}{360^0} \times 2\pi r$ T

[CBSE 2023]

A

$A = \frac{\theta}{360} \pi r^2$

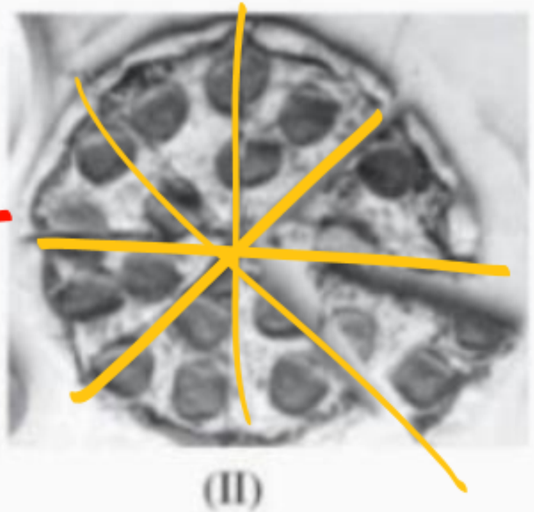
R

$l = \frac{\theta}{360} 2\pi r$

#LP:

We all love to eat pizzas, especially kids and a variety of pizzas are available in India which have been modified according to Indian taste and menu. From the Greeks to the Egyptians, from the Persians to the Indians, there have been incarnations of pizza served throughout history, Flatbreads, naan and plakountos are all early preparations that could be considered cousins to the modern pizza, but there isn't a consensus as to which is first and whether these could even be considered precursors to pizza at all. Consider two pizzas, both of equal diameter, namely, 12 inches. The first pizza marked (I) has been cut into six equal slices, whereas the second pizza, marked (II) has been cut into eight equal slices.

(I)



(II)

Based on the above information, solve the following questions:

Q1. The area of one slice in pizza, marked (I) is:

a. 6π sq. inches

b. 8π sq. inches

c. 10π sq. inches

d. None of these

Q2. The perimeter of the pizza slice shown in (1) is:

a. $(\pi + 12)$ inch

b. $(\pi + 10)$ inch

c. $(2\pi + 10)$ inch

d. $(2\pi + 12)$ inch

Q3. The ratio of area of slice to the area of remaining pizza in (1) is:

a. 5:1

b. 1:5

c. 2:5

d. 5:3

Q4. The ratio of areas of each slice of pizza (1) and (II) is:

a. 3:4

b. 5:3

c. 4:3

d. 2:5

Q5. The relation between area of a sector (A), length of the arc (l), angle (θ) subtended by the arc at the centre and radius of circle is:

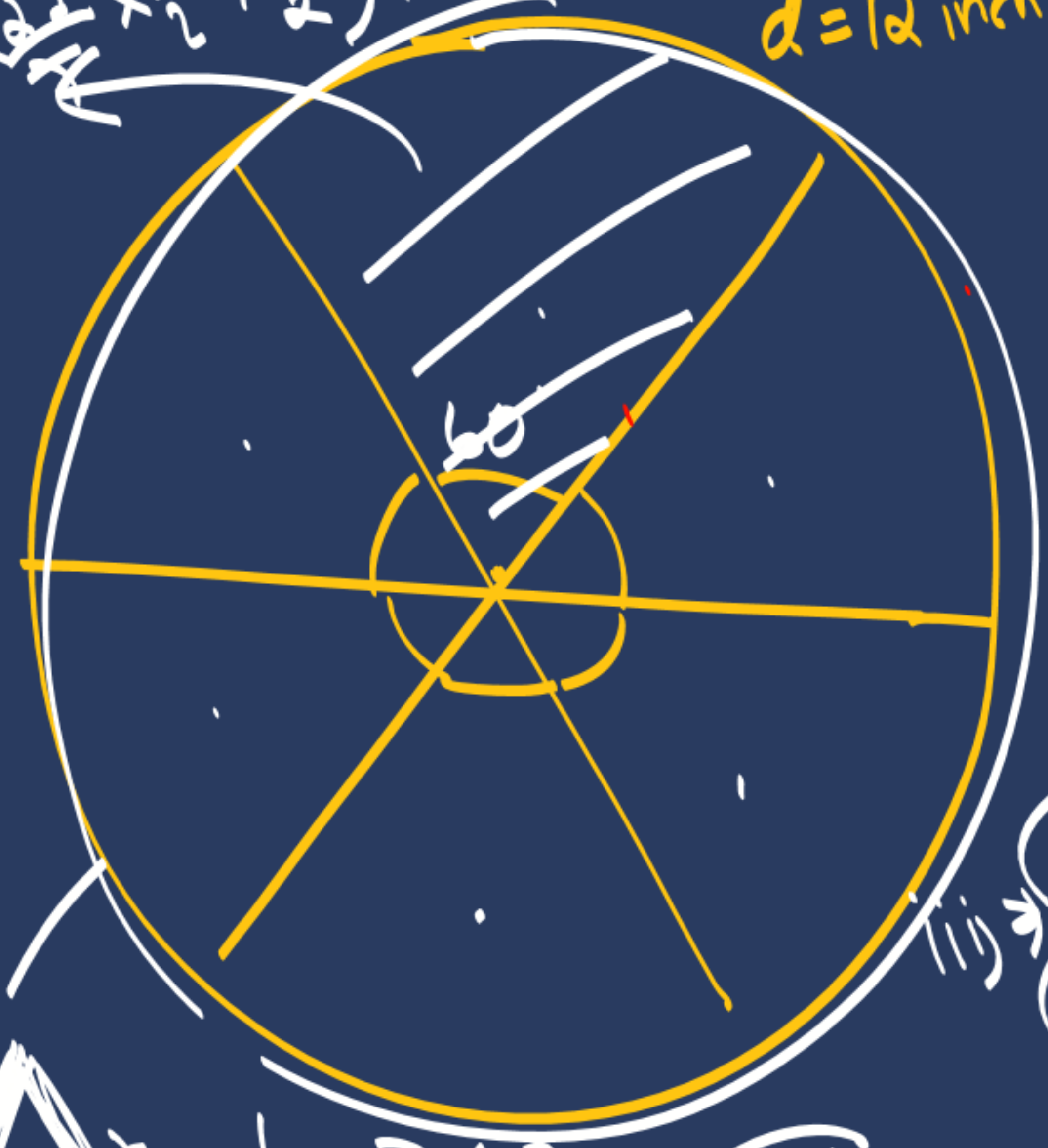
a. $\frac{1}{2}lr$

b. lr

c. $\frac{1}{3}lr$

d. $\frac{1}{2}lr^2$

$A = \left(\frac{60}{360} \times \frac{22}{7} \times \frac{12}{2} \times \frac{12}{2}\right) \text{ inch}^2$

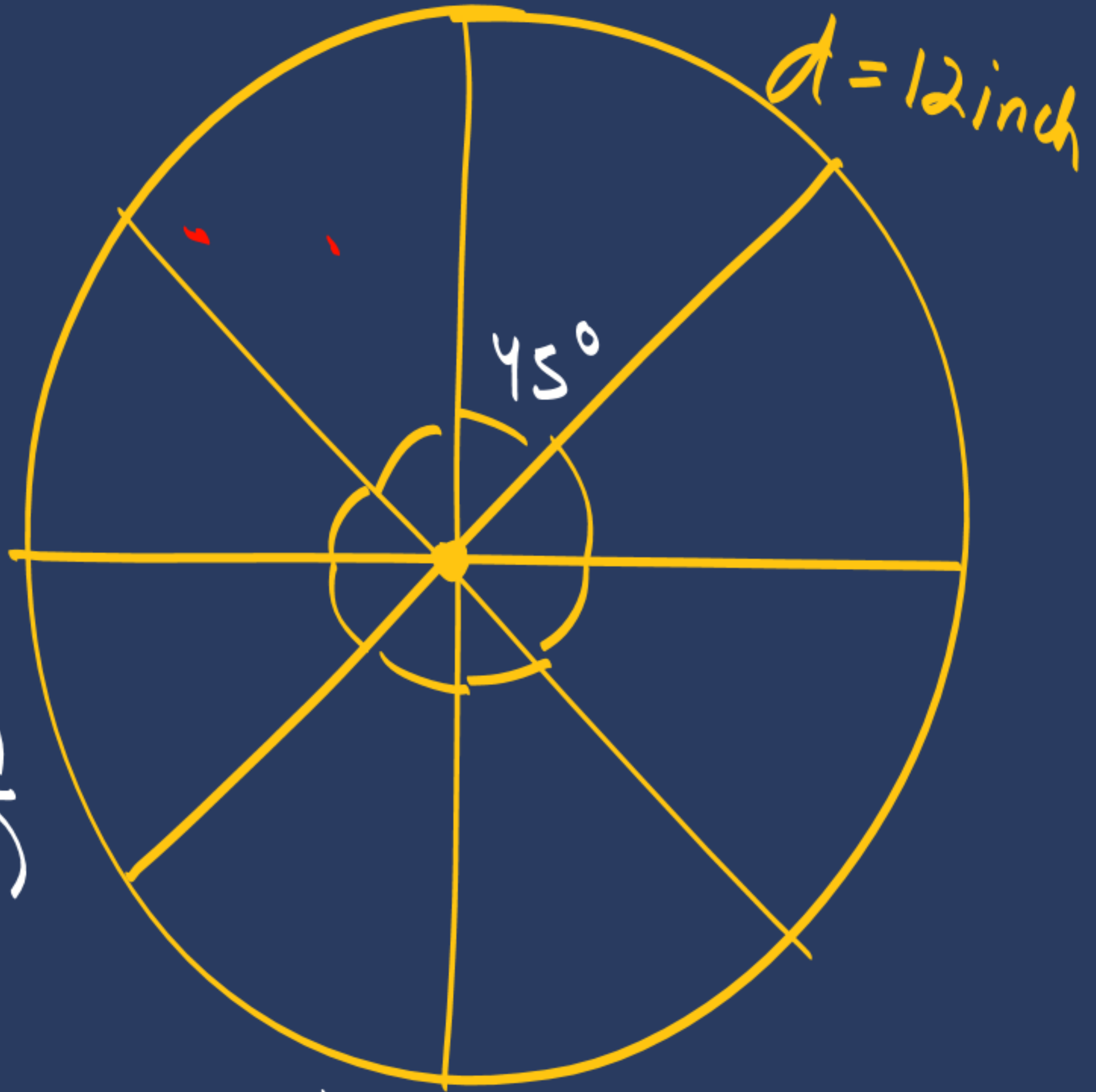


(ii) → $\frac{(\text{Slice area})}{(\text{rem. pizza})}$

$\frac{360}{6} \rightarrow 60°$

(i)

$P = \left[r + r + \frac{\theta}{360} 2\pi r\right]$



$\frac{360}{8} \rightarrow 45°$

(ii) $\dfrac{\text{ar. sector 1}}{\text{ar sector 2}} = \dfrac{\frac{\theta_1}{\cancel{360}} \times \cancel{\pi r^2}}{\frac{\theta_2}{\cancel{360}} \times \cancel{\pi r^2}}$

$= \dfrac{\theta_1}{\theta_2} = \dfrac{\cancel{60}\ \cancel{12}\ 4}{\cancel{45}\ \cancel{9}\ 3} = \dfrac{4}{3}$

(v) $\dfrac{A = \frac{\theta}{360} \pi r^2}{l = \frac{\theta}{360} 2\pi r} \Rightarrow \dfrac{A}{l} = \dfrac{r}{2}$

$A = \dfrac{rl}{2}$

#LP: The Olympic symbol, comprising five interlocking rings, represents the union of the five continents of the world and the meeting of athletes from all over the world at the Olympic Games. In order to spread awareness about the Olympic Games, students of Class X took part in various activities organised by the school. One such group of students made 5 circular rings in the school lawn with the help of ropes. Each circular ring required 44 m of rope.

Also, in the shaded regions, as shown in the figure, students made rangoli showcasing various sports and games. It is given that OAB is an equilateral triangle, and all unshaded regions are congruent.

Based on the above information, answer the following questions:

i. Find the radius of each circular ring. (1)

$2\pi r = 44m$

$2\times\frac{22}{7}\times r = 44$

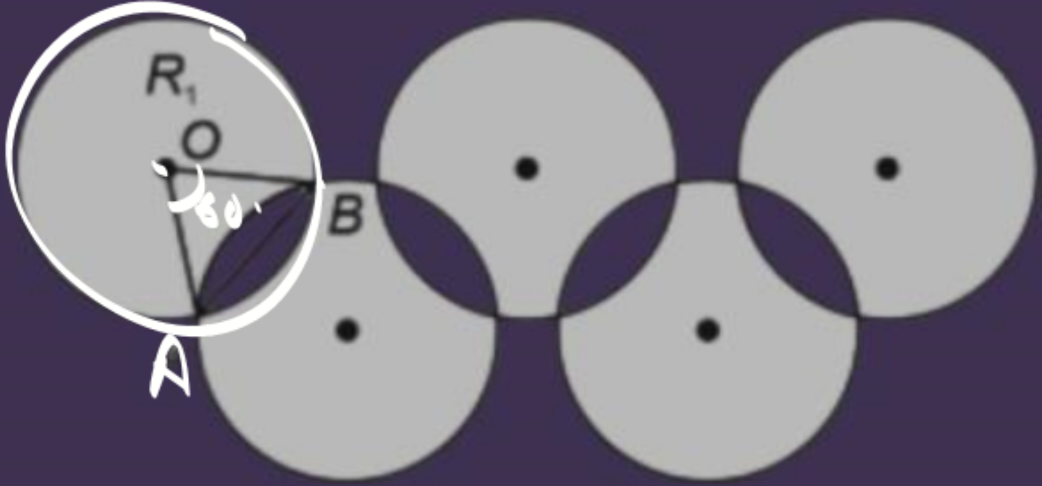
$r = 7m$

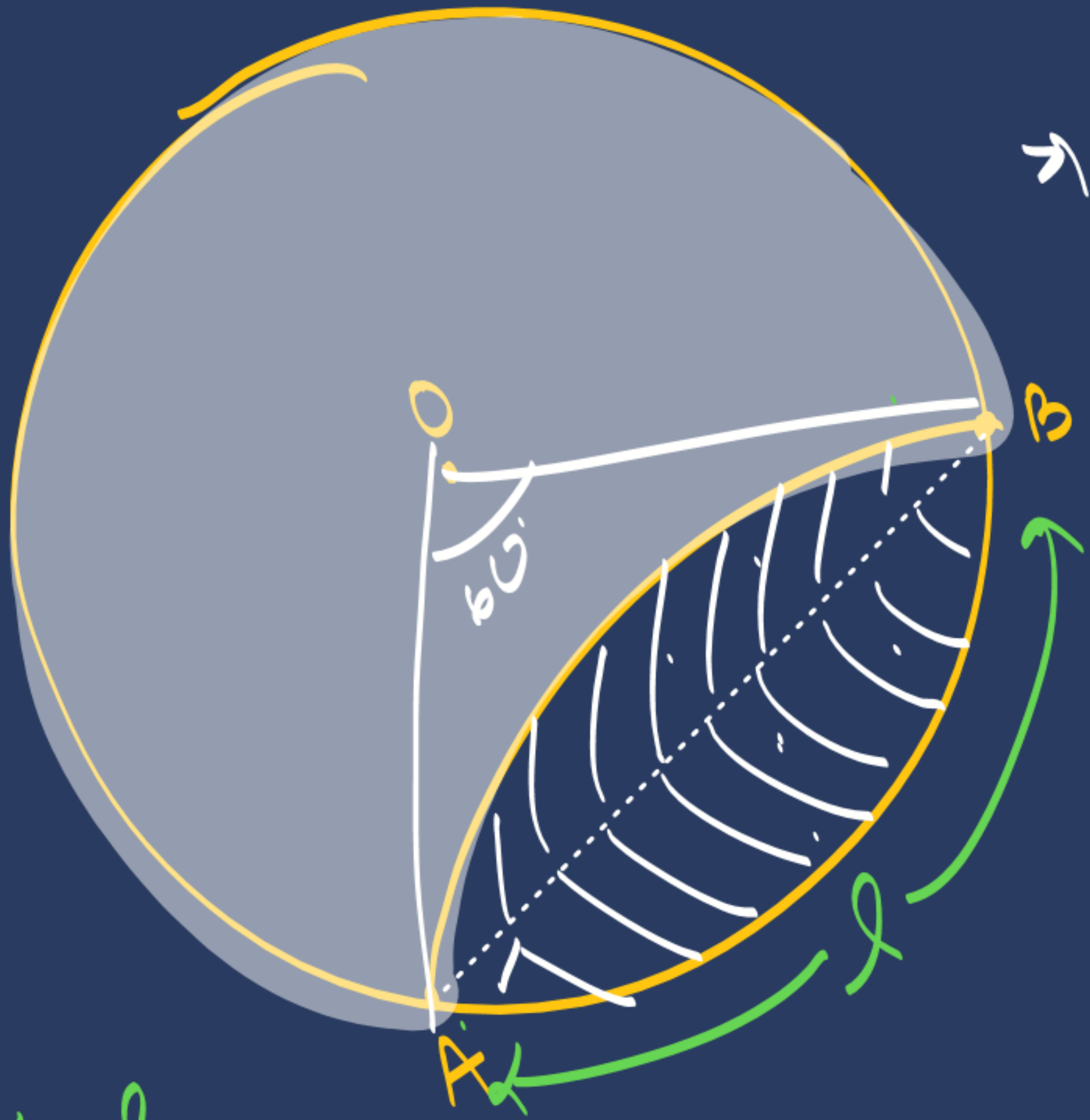
ii. What is the measure of angle AOB? (1) 60°

iii. a. Find the area of the shaded region R1. (2)

OR

b. Find the length of rope around the unshaded regions





$\rightarrow$ ar. Circle $-$ (2 × ar. segment)

$\rightarrow \pi r^2 - 2\left[\text{sector} - \Delta OAB\right]$

$\rightarrow \pi r^2 - 2\left(\frac{60}{360} \times \pi r^2 - \frac{\sqrt{3}}{4} r^2\right)$

(b) $2 \times l$

$\rightarrow \left(2 \times \frac{60}{360} \times 2\pi r\right)$

Next Toppers

5/14 completed !

Thank you Coodiess !!

$\left.\begin{matrix}6\\7\\8\end{matrix}\right] \rightarrow$ 5pm